

# Notes on Thermophoretic Force: a Mathematica verification

Reproduction and check of C. Chin, "Notes on Thermophoretic Force" (ThermophoreticForce\_v21.pdf, February 2026)

This notebook reproduces the notes section by section: the text, the equations with their original numbers, and the tables. After each claim that can be checked, a shaded **Verification** block tests it with symbolic algebra, numerical integration, unit analysis or an independent kinetic-theory derivation. Every test ends in a badge:

**VERIFIED** means the statement in the notes holds as written. **DISCREPANCY** means it does not, and the block shows the corrected form. **NOTE** marks a remark that is not a pass/fail test, such as a hidden assumption or a range of validity. Section 7 collects every result in one table.

The text in the unshaded cells follows the original notes. Text in shaded cells, and all code, belongs to the verification. Equation numbers in the right margin are those of the PDF.

## 0. Verification toolkit

Helper definitions used throughout: the hat (skew) map, the quaternion rotation matrix and kinematic matrix of Section 1.5, symmetry-group generators, and the bookkeeping for verification results.

```
ClearAll["Global`*"];
$verifications = {};
badge[status_] :=
  Switch[status, "VERIFIED", Style["✓ VERIFIED", Bold, RGBColor[0.1, 0.5, 0.15]],
    "DISCREPANCY", Style["△ DISCREPANCY", Bold, RGBColor[0.75, 0.1, 0.1]],
    "NOTE", Style["● NOTE", Bold, RGBColor[0.15, 0.35, 0.75]]];
record[ref_, claim_, status_, detail_] :=
  (AppendTo[$verifications, {ref, claim, status, detail}];
   VR[ref, claim, status, detail]);
VR /: MakeBoxes[VR[ref_, claim_, status_, detail_, fmt_] := ToBoxes[
  Panel[Column[{Row[{badge[status], " ", Style[ref, Bold], ": ", claim}], If[
    detail == "", Nothing, Style[detail, 11, GrayLevel[0.25]]}], Spacings → 0.6],
    Background → Switch[status, "VERIFIED", RGBColor[0.92, 0.97, 0.92],
      "DISCREPANCY", RGBColor[0.99, 0.92, 0.92], _, RGBColor[0.92, 0.94, 0.99]],
    FrameMargins → 10, ImageSize → Full], fmt];
```

```

verify[ref_String, claim_String, test_, detail_ : "" ] :=
  record[ref, claim, If[TrueQ[test], "VERIFIED", "DISCREPANCY"], detail];
note[ref_String, claim_String, detail_ : "" ] := record[ref, claim, "NOTE", detail];
sci[x_] := If[x == 0, "0", Module[{e = Floor[Log10[Abs[N[x]]]}],
  ToString[NumberForm[N[x] / 10^e, {3, 2}]] <> "e" <> ToString[e]]];
num[x_, n_ : 3] :=
  If[x == 0, "0", TextString[N[Round[x, 10.^ (Floor[Log10[Abs[N[x]]] - n + 1)]]]];
zeroQ[expr_] := AllTrue[Flatten[{Simplify[expr]}], PossibleZeroQ];
hat[{a1_, a2_, a3_}] := {{0, -a3, a2}, {a3, 0, -a1}, {-a2, a1, 0}};
vecOfHat[m_] := {m[[3, 2]], m[[1, 3]], m[[2, 1]]};
quatR[{q0_, q1_, q2_, q3_}] := (q0^2 - {q1, q2, q3} . {q1, q2, q3}) IdentityMatrix[3] +
  2 Outer[Times, {q1, q2, q3}, {q1, q2, q3}] + 2 q0 hat[{q1, q2, q3}];
Qmat[{q0_, q1_, q2_, q3_}] :=
  {{-q1, -q2, -q3}, {q0, -q3, q2}, {q3, q0, -q1}, {-q2, q1, q0}};
axisAngleR[k_, psi_] :=
  Cos[psi] IdentityMatrix[3] + (1 - Cos[psi]) Outer[Times, k, k] + Sin[psi] hat[k];
symMat[s_] := Table[s[Min[i, j], Max[i, j]], {i, 3}, {j, 3}];
genMat[s_] := Table[s[i, j], {i, 3}, {j, 3}];
Zhat = {0, 0, 1};
greekRules = Join[{
  "Sqrt[p_i]" -> "\sqrt{\pi}", "Sqrt[\Pi]" -> "\sqrt{\pi}", "grad " -> "\nabla",
  Map[RegularExpression["\\b" <> #[[1]] <> "(?! [A-Za-z])" -> #[[2]] &,
    {"alpha", "\alpha"}, {"gamma", "\gamma"}, {"delta", "\delta"}, {"epsilon", "\epsilon"}, {"eps", "\epsilon"},
    {"eta", "\eta"}, {"theta", "\theta"}, {"kappa", "\kappa"}, {"lambda", "\lambda"}, {"mu", "\mu"},
    {"nu", "\nu"}, {"xi", "\xi"}, {"pi", "\pi"}, {"rho", "\rho"}, {"sigma", "\sigma"},
    {"tau", "\tau"}, {"phi", "\phi"}, {"psi", "\psi"}, {"omega", "\omega"}, {"Gamma", "\Gamma"},
    {"Delta", "\Delta"}, {"Lambda", "\Lambda"}, {"Omega", "\Omega"}, {"Pi", "\Pi"}}];
prettyRow[s0_String] :=
  Module[{s = StringReplace[s0, greekRules]}, Row[StringSplit[s,
    {RegularExpression["([A-Za-z\\x{0391}-\\x{03C9}]+) _\\{([^}]*)\\}" ->
      Subscript[Style["$1", Italic], "$2"], RegularExpression[
        "([A-Za-z\\x{0391}-\\x{03C9}]+) _([A-Za-z0-9\\x{0391}-\\x{03C9}]+)" ->
          Subscript[Style["$1", Italic], "$2"],
        RegularExpression["([A-Za-z\\x{0391}-\\x{03C9}])\\^([A-Za-z0-9]+)" ->
          Superscript["$1", "$2"]}]]];
Print["Toolkit loaded: ", $Version];
Toolkit loaded: 15.0.1 for Mac OS X ARM (64-bit) (July 2, 2026)

```

## 1. General formalism for rigid-body motion in a thermal gas cell

The disk and cylinder examples in Sections 2 and 3 exploit specific geometric symmetries to simplify the force expressions. In this section we derive the equations of motion for a rigid body of completely

arbitrary shape, making no symmetry assumption. The only assumption throughout Sections 1.1–1.6 is that gravity acts along  $-\hat{\mathbf{Z}}$ . The temperature gradient  $\nabla T$  is treated as a general vector with arbitrary direction. Symmetric bodies such as cylinders and spheroids emerge as special cases of this formulation, which are discussed in Section 1.7.

## 1.1 Reference frames, rotation matrix, and inertia tensor

**Lab frame.** We work in a fixed inertial frame ( $\hat{\mathbf{X}}, \hat{\mathbf{Y}}, \hat{\mathbf{Z}}$ ). The only assumption placed on the external fields is that gravity acts along  $-\hat{\mathbf{Z}}$ :

$$\mathbf{g} = -g\hat{\mathbf{Z}}. \quad (1)$$

The macroscopic temperature gradient  $\nabla T$  is a prescribed vector field in the lab frame. No assumption is made about its direction.

**Body frame and inertia tensor.** We attach a frame ( $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$ ) to the rigid body with its origin at the centre of mass (CoM), aligned with the principal axes of inertia so that the inertia tensor is diagonal:

$$\mathbf{I} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}, \quad (2)$$

where  $I_1, I_2, I_3$  are the three principal moments of inertia. In general all three are distinct,  $I_1 \neq I_2 \neq I_3$ .

**Rotation matrix.** The instantaneous orientation of the body is described by the rotation matrix  $R \in \text{SO}(3)$ , defined so that a vector  $\mathbf{u}$  with components in the body frame has lab-frame components  $\mathbf{u} = R\mathbf{u}^b$ , with inverse  $\mathbf{u}^b = R^T\mathbf{u}$ . In particular, the temperature gradient and the CoM velocity in the body frame are

$$(\nabla T)^b = R^T \nabla T, \quad \mathbf{v}^b = R^T \mathbf{v}, \quad (3)$$

where  $\mathbf{v} = \dot{\mathbf{x}}$  is the lab-frame CoM velocity. The evolution of  $R$  in time is governed by the quaternion kinematics derived in Section 1.5.

**Verification of (3).** The inverse  $\mathbf{u}^b = R^T\mathbf{u}$  requires  $R^T R = \mathbf{I}_3$ . We check this, and  $\det R = 1$ , for the general unit-quaternion parametrisation (18) used later.

```
Rq = quatR[{q0, q1, q2, q3}];
unit = q0^2 + q1^2 + q2^2 + q3^2 == 1;
verify["Eq. (3)",
  "R(q) is orthogonal with det R = 1, so u^b = R^T u inverts u = R u^b",
  Simplify[Transpose[Rq].Rq, unit] === IdentityMatrix[3] &&
  Simplify[Det[Rq], unit] === 1]
```

✓ **VERIFIED** Eq. (3):  $R(q)$  is orthogonal with  $\det R = 1$ , so  $\mathbf{u}^b = R^T \mathbf{u}$  inverts  $\mathbf{u} = R \mathbf{u}^b$

## 1.2 Gas-interaction tensors for an arbitrary shape

In the low-velocity, small-temperature-gradient limit all forces and torques respond linearly to their driving fields. We introduce four tensors in the natural order: the two forces (from temperature gradient and from velocity) followed by the two torques (from temperature gradient and from angular velocity).

**Thermophoretic force tensor  $\mathbf{f}_{\text{th}}$  (symmetric, 6 DOF).** Thermophoresis drives particles from hot toward cold. The tensor  $\mathbf{f}_{\text{th}}$  relates the temperature gradient to the thermophoretic force in the body frame:

$$\mathbf{F}_{\text{th}}^b = -\mathbf{f}_{\text{th}} (\nabla \ln T)^b, \quad \mathbf{f}_{\text{th}} = \begin{pmatrix} f_{11} & f_{12} & f_{13} \\ f_{12} & f_{22} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix}. \quad (4)$$

$\mathbf{f}_{\text{th}}$  is symmetric positive-definite (6 DOF). The off-diagonal entry  $f_{ij}$  means a gradient along  $\hat{\mathbf{e}}_j$  drives a force component along  $\hat{\mathbf{e}}_i$ .

**Drag force tensor  $\mathbf{f}_{\text{drag}}$  (symmetric, 6 DOF).** The tensor  $\mathbf{f}_{\text{drag}}$  relates the CoM velocity to the drag force in the body frame:

$$\mathbf{F}_{\text{drag}}^b = -\mathbf{f}_{\text{drag}} \mathbf{v}^b, \quad \mathbf{f}_{\text{drag}} = \begin{pmatrix} d_{11} & d_{12} & d_{13} \\ d_{12} & d_{22} & d_{23} \\ d_{13} & d_{23} & d_{33} \end{pmatrix}. \quad (5)$$

$\mathbf{f}_{\text{drag}}$  is symmetric positive-definite (6 DOF): drag power  $P = -\mathbf{v}^T \mathbf{f}_{\text{drag}} \mathbf{v} < 0$  for all  $\mathbf{v}$  forces positive-definiteness, and only the symmetric part contributes to  $P$ .

**Verification of the dissipation argument.** Only the symmetric part of a matrix contributes to the quadratic form  $\mathbf{v}^T \mathbf{A} \mathbf{v}$ , because the antisymmetric part gives zero identically.  $P < 0$  for all  $\mathbf{v}$  is then exactly the statement that the symmetric part is positive-definite. The symmetry of  $\mathbf{f}_{\text{drag}}$  itself does not follow from dissipation; it follows from the Lorentz/Onsager reciprocal relations. The same remark applies to  $\mathbf{f}_{\text{th}}$ , which is *assumed* symmetric here.

```
vv = {v1, v2, v3};
Aanti = {{0, a12, a13}, {-a12, 0, a23}, {-a13, -a23, 0}};
Column[{verify["Eq. (5)",
  "antisymmetric part of the drag tensor does no work (v.A.v = 0)",
  zeroQ[vv.Aanti.vv]], note["Eqs. (4)-(5)",
  "symmetry of f_drag rests on reciprocity; symmetry of f_th is an assumption",
  "Dissipation only constrains the symmetric part. No symmetry
    principle makes f_th symmetric for a generic (chiral)
    body; see the symmetry analysis in Section 1.7, where the
    chiral group C_inf allows an antisymmetric part of f_th."]]}]
```

✓ **VERIFIED** Eq. (5): antisymmetric part of the drag tensor does no work ( $\mathbf{v} \cdot \mathbf{A} \cdot \mathbf{v} = 0$ )

● **NOTE Eqs. (4)-(5):** symmetry of  $f_{\text{drag}}$  rests on reciprocity; symmetry of  $f_{\text{th}}$  is an assumption. Dissipation only constrains the symmetric part. No symmetry principle makes  $f_{\text{th}}$  symmetric for a generic (chiral) body; see the symmetry analysis in Section 1.7, where the chiral group  $C_{\text{inf}}$  allows an antisymmetric part of  $f_{\text{th}}$ .

**Thermophoretic torque tensor  $T_{\text{th}}$  (9 DOF).** The temperature gradient can exert a net torque on a non-symmetric body:

$$\tau_{\text{th}}^b = -T_{\text{th}} (\nabla \ln T)^b, \quad T_{\text{th}} = \begin{pmatrix} t_{11} & t_{12} & t_{13} \\ t_{21} & t_{22} & t_{23} \\ t_{31} & t_{32} & t_{33} \end{pmatrix}. \quad (6)$$

$T_{\text{th}}$  is in general a full  $3 \times 3$  matrix (9 DOF) with no symmetry required. For bodies with three mutually perpendicular mirror planes  $T_{\text{th}} = 0$ .

**Rotational drag tensor  $T_{\text{drag}}$  (symmetric, 6 DOF).** The tensor  $T_{\text{drag}}$  relates angular velocity to the resistive torque in the body frame:

$$\tau_{\text{drag}}^b = -T_{\text{drag}} \omega^b, \quad T_{\text{drag}} = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{12} & r_{22} & r_{23} \\ r_{13} & r_{23} & r_{33} \end{pmatrix}. \quad (7)$$

$T_{\text{drag}}$  is symmetric positive-definite (6 DOF) by the same dissipation argument as  $f_{\text{drag}}$ .

**Verification of " $T_{\text{th}} = 0$  for three mirror planes".**  $T_{\text{th}}$  maps a polar vector  $(\nabla \ln T)$  to an axial vector (torque), so it transforms as a pseudotensor,  $T \rightarrow \det(g) g T g^T$ . Invariance under the three reflections  $x \rightarrow -x, y \rightarrow -y, z \rightarrow -z$  forces every entry to vanish. (Section 1.7 repeats this analysis for every symmetry class.)

```
mirror[n_] := IdentityMatrix[3] - 2 Outer[Times, n, n] / (n.n);
pseudoInvariantSolution[gens_] := Module[{X = genMat[t], eqs},
  eqs = Flatten[Table[X - Det[gm] gm.X.Transpose[gm], {gm, gens}]];
  X /. First@Solve[Thread[eqs == 0], Flatten[X]];
TthD2h =
  pseudoInvariantSolution[{mirror[{1, 0, 0}], mirror[{0, 1, 0}], mirror[{0, 0, 1}]}];
verify["Sec. 1.2", "three mutually perpendicular mirror planes force T_th = 0",
  TthD2h === ConstantArray[0, {3, 3}]]
```

✓ **VERIFIED Sec. 1.2:** three mutually perpendicular mirror planes force  $T_{\text{th}} = 0$

### 1.3 Force and torque summary

The total force and torque on the body in the body frame are:

$$\mathbf{F}^b = \mathbf{F}_{\text{th}}^b + \mathbf{F}_{\text{drag}}^b = -\mathbf{f}_{\text{th}} (\nabla \ln T)^b - \mathbf{f}_{\text{drag}} \mathbf{v}^b, \quad (8)$$

$$\boldsymbol{\tau}^b = \boldsymbol{\tau}_{\text{th}}^b + \boldsymbol{\tau}_{\text{drag}}^b = -\mathbf{T}_{\text{th}} (\nabla \ln T)^b - \mathbf{T}_{\text{drag}} \boldsymbol{\omega}^b. \quad (9)$$

These four tensors and their driving fields can be collected into a single  $6 \times 6$  linear response:

$$\begin{pmatrix} \mathbf{F}^b \\ \boldsymbol{\tau}^b \end{pmatrix} = - \begin{pmatrix} \mathbf{f}_{\text{th}} & \mathbf{f}_{\text{drag}} \\ \mathbf{T}_{\text{th}} & \mathbf{T}_{\text{drag}} \end{pmatrix} \begin{pmatrix} (\nabla \ln T)^b \\ \mathbf{v}^b \end{pmatrix}. \quad (10)$$

Note that the driving fields are  $\nabla \ln T$  and  $\mathbf{v}$  (not  $\boldsymbol{\omega}$ ); the rotational responses  $\tau_{\text{drag}}$  and the translational drag  $F_{\text{drag}}$  both arise from velocity fields in the gas. The four blocks have different physical dimensions:

$$[\mathbf{f}_{\text{th}}] = F \cdot L, \quad [\mathbf{f}_{\text{drag}}] = F \cdot T/L, \quad [\mathbf{T}_{\text{th}}] = F \cdot L^2, \quad [\mathbf{T}_{\text{drag}}] = F \cdot L^2 \cdot T/L. \quad (11)$$

The total DOF for a body of arbitrary shape is  $6 + 6 + 9 + 6 = 27$  (plus 6 for the inertia tensor). See Section 1.7 for how symmetry reduces this.

**Verification of (10) against (8)–(9).** Multiply out the block matrix of (10). The force row reproduces (8). The torque row, however, gives  $-\mathbf{T}_{\text{th}}(\nabla \ln T)^b - \mathbf{T}_{\text{drag}} \mathbf{v}^b$ , i.e. rotational drag driven by the *linear* velocity, whereas (7) and (9) have  $\mathbf{T}_{\text{drag}} \boldsymbol{\omega}^b$ . The two are not even dimensionally compatible:  $\mathbf{T}_{\text{drag}} \mathbf{v}$  is not a torque. The consistent form needs three driving fields. The general linear response additionally has translation–rotation coupling blocks, which (8)–(9) omit.

```
fth = symMat[f]; fdr = symMat[d]; Tth = genMat[t]; Tdr = symMat[r];
gb = {g1, g2, g3}; vb = {vb1, vb2, vb3}; wb = {w1, w2, w3};
blockEq10 = -ArrayFlatten[{{fth, fdr}, {Tth, Tdr}}].Join[gb, vb];
Column[{verify["Eq. (10)", "force rows of (10) reproduce (8)",
  zeroQ[blockEq10[[1 ;; 3]] - (-fth.gb - fdr.vb)]],
  verify["Eq. (10)", "torque rows of (10) reproduce (9)",
  zeroQ[blockEq10[[4 ;; 6]] - (-Tth.gb - Tdr.wb)], "The torque rows give -T_th g -
  T_drag v^b instead of -T_th g - T_drag w^b. The text below (10), 'the
  driving fields are grad ln T and v (not w)', repeats the slip."]]]
```

✓ **VERIFIED** Eq. (10): force rows of (10) reproduce (8)

⚠ **DISCREPANCY** Eq. (10): torque rows of (10) reproduce (9)

The torque rows give  $-\mathbf{T}_{\text{th}} \mathbf{g} - \mathbf{T}_{\text{drag}} \mathbf{v}^b$  instead of  $-\mathbf{T}_{\text{th}} \mathbf{g} - \mathbf{T}_{\text{drag}} \boldsymbol{\omega}^b$ . The text below (10), 'the driving fields are  $\nabla \ln T$  and  $\mathbf{v}$  (not  $\mathbf{w}$ )', repeats the slip.

The consistent collection of (8)–(9) uses three driving fields:

$$\begin{pmatrix} \mathbf{F}^b \\ \boldsymbol{\tau}^b \end{pmatrix} = - \begin{pmatrix} \mathbf{f}_{\text{th}} & \mathbf{f}_{\text{drag}} & 0 \\ \mathbf{T}_{\text{th}} & 0 & \mathbf{T}_{\text{drag}} \end{pmatrix} \begin{pmatrix} (\nabla \ln T)^b \\ \mathbf{v}^b \\ \boldsymbol{\omega}^b \end{pmatrix}$$

```

units = <|"F" → Quantity[1, "Newtons"],
  "L" → Quantity[1, "Meters"], "T" → Quantity[1, "Seconds"] |>;
gradLnT = Quantity[1, 1 / "Meters"];
vel = Quantity[1, "Meters" / "Seconds"];
omega = Quantity[1, 1 / "Seconds"];
dimOK[coef_, driving_, response_] := CompatibleUnitQ[coef driving, response];
Column[{verify["Eq. (11)", "[f_th] = F L: f_th times grad ln T is a force",
  dimOK[units["F"] units["L"], gradLnT, units["F"]]],
  verify["Eq. (11)", "[f_drag] = F T/L: f_drag times v is a force",
  dimOK[units["F"] units["T"] / units["L"], vel, units["F"]]],
  verify["Eq. (11)", "[T_th] = F L^2: T_th times grad ln T is a torque",
  dimOK[units["F"] units["L"]^2, gradLnT, units["F"] units["L"]]], verify[
  "Eq. (11)", "[T_drag] = F L^2 T/L (= F L T): T_drag times omega is a torque",
  dimOK[units["F"] units["L"]^2 units["T"] / units["L"],
  omega, units["F"] units["L"]]], verify["Sec. 1.3",
  "DOF count 6 + 6 + 9 + 6 = 27", Length[Variables[{fth, fdr, Tth, Tdr}]] == 27]]]

```

✓ **VERIFIED** Eq. (11):  $[f_{th}] = F L$ :  $f_{th}$  times  $\nabla \ln T$  is a force

✓ **VERIFIED** Eq. (11):  $[f_{drag}] = F T/L$ :  $f_{drag}$  times  $v$  is a force

✓ **VERIFIED** Eq. (11):  $[T_{th}] = F L^2$ :  $T_{th}$  times  $\nabla \ln T$  is a torque

✓ **VERIFIED** Eq. (11):  $[T_{drag}] = F L^2 T/L (= F L T)$ :  $T_{drag}$  times  $\omega$  is a torque

✓ **VERIFIED** Sec. 1.3: DOF count  $6 + 6 + 9 + 6 = 27$

## 1.4 Equations of motion

Substituting the forces (8) and torques (9) into Newton's second law and Euler's equation, and expressing  $(\nabla \ln T)^b = R^T \nabla \ln T$ ,  $\mathbf{v}^b = R^T \mathbf{v}$ , the complete equations of motion for the state  $(\mathbf{x}, \mathbf{v}, R, \boldsymbol{\omega}^b)$  are:

$$\dot{\mathbf{x}} = \mathbf{v}, \quad (12)$$

$$M \dot{\mathbf{v}} = -M g \hat{\mathbf{z}} - R[\mathbf{f}_{th} R^T \nabla \ln T + \mathbf{f}_{drag} R^T \mathbf{v}], \quad (13)$$

$$\dot{R} = R \hat{\boldsymbol{\omega}}^b, \quad (14)$$

$$I \dot{\boldsymbol{\omega}}^b = -\boldsymbol{\omega}^b \times (I \boldsymbol{\omega}^b) - \mathbf{T}_{th} R^T \nabla \ln T - \mathbf{T}_{drag} \boldsymbol{\omega}^b. \quad (15)$$

Here  $\hat{\mathbf{a}}$  denotes the  $3 \times 3$  skew-symmetric matrix associated with a vector  $\mathbf{a}$ , defined by  $\hat{\mathbf{a}} \mathbf{u} = \mathbf{a} \times \mathbf{u}$  for all  $\mathbf{u}$ , with entries  $\hat{a}_{ij} = -\varepsilon_{ijk} a_k$ . In (13) gravity acts downward ( $-\hat{\mathbf{z}}$ ) and the temperature gradient  $\nabla T$  is also taken along  $-\hat{\mathbf{z}}$  (hot above cold), so the thermophoretic force  $-\mathbf{f}_{th} R^T \nabla \ln T$  points upward and can levitate the particle. Equation (14) is equivalent to the lab-frame form  $\dot{R} = \hat{\boldsymbol{\omega}}^{lab} R$  with  $\boldsymbol{\omega}^{lab} = R \boldsymbol{\omega}^b$ .

**Verification of Section 1.4.** (i) Rotating the body-frame force (8) into the lab and adding gravity gives (13). (ii) The hat map has entries  $-\varepsilon_{ijk} a_k$ . (iii)  $R \hat{\boldsymbol{\omega}}^b R^T = R \hat{\boldsymbol{\omega}}^b$  for any rotation, which proves the

equivalence of the body and lab forms of (14). (iv) The direction statement: with  $\nabla T = -|\nabla T|\hat{Z}$  the temperature *decreases* with height, so the gas is hot *below* and cold above, and the force on a sphere points up.

```
lnTlab = {gx, gy, gz};
Fb8 = -fth.(Transpose[Rq].lnTlab) - fdr.(Transpose[Rq].{vx, vy, vz});
Column[{verify["Eq. (13)",
  "M dv/dt = -M g Z - R[f_th R^T grad ln T + f_drag R^T v] follows from (8)",
  zeroQ[(-Mm gg Zhat + Rq.Fb8) -
    (-Mm gg Zhat - Rq.(fth.Transpose[Rq].lnTlab + fdr.Transpose[Rq].{vx, vy, vz}))]],
  verify["Sec. 1.4", "hat map entries are -epsilon_ijk a_k", hat[{a1, a2, a3}] ==
    Table[-Sum[LeviCivitaTensor[3][[i, j, k]] {a1, a2, a3}][[k]], {k, 3}],
    {i, 3}, {j, 3}]], verify["Eq. (14)",
  "R hat(w^b) R^T = hat(R w^b), so dR/dt = R hat(w^b) = hat(w^lab) R",
  Simplify[Rq.hat[wb].Transpose[Rq] - hat[Rq.wb], unit] ==
    ConstantArray[0, {3, 3}]],
  verify["Sec. 1.4", "'grad T along -Z (hot above cold)'", False,
  "With grad T = -|grad T| Z the temperature falls with height: hot
    below, cold above. The conclusion that the thermophoretic
    force points up (+Z, toward cold) is correct: -f0 R^T
    grad ln T = +f0 |grad ln T| Z for an isotropic f_th."]]]
```

✓ **VERIFIED** Eq. (13):  $M dv/dt = -M g Z - R[f_{th} R^T \nabla \ln T + f_{drag} R^T v]$  follows from (8)

✓ **VERIFIED** Sec. 1.4: hat map entries are  $-\epsilon_{ijk} a_k$

✓ **VERIFIED** Eq. (14):  $R \text{hat}(w^b) R^T = \text{hat}(R w^b)$ , so  $dR/dt = R \text{hat}(w^b) = \text{hat}(w^{lab}) R$

⚠ **DISCREPANCY** Sec. 1.4: ' $\nabla T$  along -Z (hot above cold)'

With  $\nabla T = -|\nabla T| Z$  the temperature falls with height: hot below, cold above. The conclusion that the thermophoretic force points up (+Z, toward cold) is correct:  $-f_0 R^T \nabla \ln T = +f_0 |\nabla \ln T| Z$  for an isotropic  $f_{th}$ .

**Gyroscopic term.** The term  $\omega^b \times (I \omega^b)$  in Eq. (15) is the gyroscopic (Euler) term arising from the non-inertial body frame:

$$\omega^b \times (I \omega^b) = \begin{pmatrix} (I_3 - I_2) \omega_2^b \omega_3^b \\ (I_1 - I_3) \omega_3^b \omega_1^b \\ (I_2 - I_1) \omega_1^b \omega_2^b \end{pmatrix}. \quad (16)$$

It vanishes for a sphere ( $I_1 = I_2 = I_3$ ) and when  $\omega^b$  is parallel to a principal axis.

**Practical implementation.** For numerical integration,  $R \in \text{SO}(3)$  is represented by a unit quaternion  $q$  to avoid the gimbal-lock singularity of Euler angles. Equation (14) then becomes  $\dot{q} = \frac{1}{2} Q(q) \omega^b$  (see Section 1.5).

```

Iten = DiagonalMatrix[{I1, I2, I3}];
Column[
{verify["Eq. (16)", "component form of w x (I w)", zeroQ[Cross[wb, Iten.wb] -
  {(I3 - I2) w2 w3, (I1 - I3) w3 w1, (I2 - I1) w1 w2}]], verify["Eq. (16)",
  "gyroscopic term vanishes for a sphere and for w along a principal axis",
  zeroQ[Cross[wb, Iten.wb] /. {I2 → I1, I3 → I1}] &&
  AllTrue[IdentityMatrix[3], zeroQ[Cross[s #, Iten.(s #)]] &]]]}

```

✓ **VERIFIED** Eq. (16): component form of  $\mathbf{w} \times (\mathbf{I} \mathbf{w})$

✓ **VERIFIED** Eq. (16): gyroscopic term vanishes for a sphere and for  $\mathbf{w}$  along a principal axis

## 1.5 Attitude kinematics — quaternion representation

Euler angles develop a coordinate singularity (gimbal lock) at pitch  $\pm 90^\circ$ . We instead represent the orientation by a unit quaternion

$$\mathbf{q} = (q_0, q_1, q_2, q_3) \in \mathbb{R}^4, \quad |\mathbf{q}|^2 \equiv q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1. \quad (17)$$

The scalar part is  $q_0 = \cos(\psi/2)$  and the vector part is  $\vec{q} = (q_1, q_2, q_3) = \hat{k} \sin(\psi/2)$ , where  $\hat{k}$  is the unit rotation axis and  $\psi$  is the rotation angle.

**Rotation matrix from quaternion.**

$$R(\mathbf{q}) = \left( q_0^2 - |\vec{q}|^2 \right) \mathbf{I}_3 + 2 \vec{q} \vec{q}^T + 2 q_0 \hat{k}, \quad \hat{k} = \begin{pmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{pmatrix}. \quad (18)$$

**Explicit axis-angle form.**

$$R(\hat{k}, \psi) = \cos \psi \mathbf{I}_3 + (1 - \cos \psi) \hat{k} \hat{k}^T + \sin \psi \hat{k}, \quad (19)$$

with  $c \equiv \cos \psi$ ,  $s \equiv \sin \psi$ :

$$R(\hat{k}, \psi) = \begin{pmatrix} k_1^2(1-c) + c & k_1 k_2(1-c) - k_3 s & k_1 k_3(1-c) + k_2 s \\ k_1 k_2(1-c) + k_3 s & k_2^2(1-c) + c & k_2 k_3(1-c) - k_1 s \\ k_1 k_3(1-c) - k_2 s & k_2 k_3(1-c) + k_1 s & k_3^2(1-c) + c \end{pmatrix}. \quad (20)$$

**Kinematic differential equation.**

$$\dot{\mathbf{q}} = \frac{1}{2} Q(\mathbf{q}) \boldsymbol{\omega}^b, \quad Q(\mathbf{q}) = \begin{pmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{pmatrix}. \quad (21)$$

To prevent numerical drift,  $\mathbf{q}$  must be renormalised after each integration step:  $\mathbf{q} \leftarrow \mathbf{q} / |\mathbf{q}|$ .

**Verification of Section 1.5.** (i) Substituting  $q_0 = \cos(\psi/2)$ ,  $\vec{q} = \hat{k} \sin(\psi/2)$  into (18) gives (19). (ii) The explicit matrix (20) equals (19). (iii) Most importantly, the quaternion ODE (21) must reproduce the attitude equation (14). Differentiate  $R(\mathbf{q}(t))$  along  $\dot{\mathbf{q}} = \frac{1}{2} Q \boldsymbol{\omega}^b$  and compare with  $R \hat{\boldsymbol{\omega}}^b$ . (iv) The ODE preserves  $|\mathbf{q}|$  exactly, since  $\mathbf{q} \cdot Q(\mathbf{q}) \boldsymbol{\omega} = 0$ ; renormalisation only removes integrator round-off.

```

kk = {k1, k2, k3}; kunit = k1^2 + k2^2 + k3^2 == 1;
R19 = axisAngleR[kk, psi];
R20 = {{k1^2 (1 - c) + c, k1 k2 (1 - c) - k3 s, k1 k3 (1 - c) + k2 s},
      {k1 k2 (1 - c) + k3 s, k2^2 (1 - c) + c, k2 k3 (1 - c) - k1 s}, {k1 k3 (1 - c) - k2 s,
      k2 k3 (1 - c) + k1 s, k3^2 (1 - c) + c}} /. {c → Cos[psi], s → Sin[psi]};
qdotRule = Thread[{q0d, q1d, q2d, q3d} → (1/2) Qmat[{q0, q1, q2, q3}].wb];
Rdot = Sum[D[Rq, var] (var /. {q0 → q0d, q1 → q1d, q2 → q2d, q3 → q3d}),
  {var, {q0, q1, q2, q3}}] /. qdotRule;
Column[
  {verify["Eqs. (18)-(19)", "quaternion form with q0 = cos(psi/2), q = k sin(psi/2)
    equals the axis-angle form",
    zeroQ[Simplify[quatR[Join[{Cos[psi/2}], kk Sin[psi/2]]] - R19, kunit]]],
  verify["Eq. (20)", "explicit matrix (20) equals (19)", zeroQ[R20 - R19]],
  verify["Eq. (21)",
    "dq/dt = Q(q) w^b / 2 implies dR/dt = R hat(w^b), i.e. reproduces (14)",
    Simplify[Rdot - Rq.hat[wb], unit] === ConstantArray[0, {3, 3}]],
  verify["Eq. (21)", "the quaternion ODE conserves |q| exactly (q . Q(q) w = 0)",
    zeroQ[{q0, q1, q2, q3}.Qmat[{q0, q1, q2, q3}].wb]]]

```

✓ **VERIFIED** Eqs. (18)-(19): quaternion form with  $q_0 = \cos(\psi/2)$ ,  $q = k \sin(\psi/2)$  equals the axis-angle form

✓ **VERIFIED** Eq. (20): explicit matrix (20) equals (19)

✓ **VERIFIED** Eq. (21):  $dq/dt = Q(q) w^b / 2$  implies  $dR/dt = R \hat{w}^b$ , i.e. reproduces (14)

✓ **VERIFIED** Eq. (21): the quaternion ODE conserves  $|q|$  exactly ( $q \cdot Q(q) w = 0$ )

## 1.6 State-vector formulation

The complete dynamical state of the rigid body is encoded in 13 scalar variables:  $\mathbf{x}$  (3),  $\mathbf{v}$  (3),  $\mathbf{q}$  (4) and  $\boldsymbol{\omega}^b$  (3),

$$\mathbf{s} = (\mathbf{x}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega}^b) \in \mathbb{R}^{13}. \quad (22)$$

**Complete equations of motion (quaternion form).** State vector  $\mathbf{s} = (\mathbf{x}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega}^b) \in \mathbb{R}^{13}$ ; orientation  $R(\mathbf{q})$  from (18).

$$\dot{\mathbf{x}} = \mathbf{v}, \quad (23)$$

$$M \dot{\mathbf{v}} = -Mg \hat{\mathbf{z}} - R(\mathbf{q})[\mathbf{f}_{\text{th}} R(\mathbf{q})^\top \nabla \ln T + \mathbf{f}_{\text{drag}} R(\mathbf{q})^\top \mathbf{v}], \quad (24)$$

$$\dot{\mathbf{q}} = \frac{1}{2} Q(\mathbf{q}) \boldsymbol{\omega}^b, \quad (25)$$

$$I \dot{\boldsymbol{\omega}}^b = -\boldsymbol{\omega}^b \times (I \boldsymbol{\omega}^b) - \mathbf{T}_{\text{th}} R(\mathbf{q})^\top \nabla \ln T - \mathbf{T}_{\text{drag}} \boldsymbol{\omega}^b. \quad (26)$$

Renormalise after each step:  $\mathbf{q} \leftarrow \mathbf{q} / |\mathbf{q}|$ .

**Verification of Section 1.6: the equations integrate consistently.** We implement (23)–(26) directly and integrate them for a generic asymmetric body with random tensors of the classes in Section 1.2, in a uniform downward gradient. Three invariants are checked along the numerical solution:

$|\mathbf{q}| = 1$ ,  $R^T R = I_3$ , and the mechanical energy balance  
 $\frac{d}{dt} \left( \frac{1}{2} M V^2 + \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{I} \boldsymbol{\omega} + M g Z \right) = P_{\text{th}} - \mathbf{v}^b \cdot \mathbf{f}_{\text{drag}} \mathbf{v}^b - \boldsymbol{\omega}^b \cdot \mathbf{T}_{\text{drag}} \boldsymbol{\omega}^b$ . The function `eomRHS` defined here is reused in Section 2.

```
eomRHS[s_? (VectorQ[#, NumericQ] &), pars_] :=
Module[{x = s[[1 ;; 3]], v = s[[4 ;; 6]], q = s[[7 ;; 10]],
  w = s[[11 ;; 13]], R, gl = pars["gradLnT"]}, R = quatR[q / Sqrt[q.q]];
Join[v, (-pars["M"] pars["g"] Zhat -
  R. (pars["fth"].Transpose[R].gl + pars["fdrag"].Transpose[R].v)) / pars["M"],
  1 / 2 Qmat[q].w, LinearSolve[pars["I"], -Cross[w, pars["I"].w] -
  pars["Tth"].Transpose[R].gl - pars["Tdrag"].w]]];
integrateEOM[pars_, s0_List, tmax_] := Module[{y, tt, sol},
  sol = NDSolveValue[{y'[tt] == eomRHS[y[tt], pars], y[0] == N[s0]}, y,
    {tt, 0, tmax}, MaxSteps -> Infinity, PrecisionGoal -> 10, AccuracyGoal -> 12];
  sol];
stateFns[sol_] := Module[{f1, f2, f3, f4}, f1[t_?NumericQ] := sol[t][[1 ;; 3]];
  f2[t_?NumericQ] := sol[t][[4 ;; 6]];
  f3[t_?NumericQ] := sol[t][[7 ;; 10]];
  f4[t_?NumericQ] := sol[t][[11 ;; 13]];
  {f1, f2, f3, f4}];
SeedRandom[2026];
randSPD[scale_] := Module[{m = RandomReal[{-1, 1}, {3, 3}]},
  scale (IdentityMatrix[3] + 0.3 (m + Transpose[m]) / 2)];
genericPars = <|"M" -> 1., "g" -> 1., "I" -> DiagonalMatrix[{0.10, 0.13, 0.17}],
  "fth" -> randSPD[1.], "fdrag" -> randSPD[1.], "Tth" -> 0.2 RandomReal[{-1, 1}, {3, 3}],
  "Tdrag" -> randSPD[1.], "gradLnT" -> {0, 0, -1.}>;
{xs, vs, qs, ws} = stateFns[integrateEOM[genericPars,
  Join[{0, 0, 0}, {0, 0, 0}, Normalize[{1, 0.2, -0.3, 0.1}], {0, 0, 0}], 200.]];
energy[t_?NumericQ] := 0.5 genericPars["M"] vs[t].vs[t] +
  0.5 ws[t].genericPars["I"].ws[t] + genericPars["M"] genericPars["g"] xs[t][[3]];
power[t_?NumericQ] := Module[{R = quatR[Normalize[qs[t]]],
  gl = genericPars["gradLnT"], vb, g0}, vb = Transpose[R].vs[t];
  g0 = Transpose[R].gl;
  -vb.genericPars["fth"].g0 - ws[t].genericPars["Tth"].g0 -
  vb.genericPars["fdrag"].vb - ws[t].genericPars["Tdrag"].ws[t]];
work = NIntegrate[power[t], {t, 0, 200.}, MaxRecursion -> 30, PrecisionGoal -> 8];
qdev = Max[Table[Abs[Norm[qs[t]] - 1], {t, 0, 200., 0.5}]];
odev = Max[Table[Norm[Transpose[quatR[qs[t]]].quatR[qs[t]] - IdentityMatrix[3]],
  {t, 0, 200., 0.5}]]];
```

```

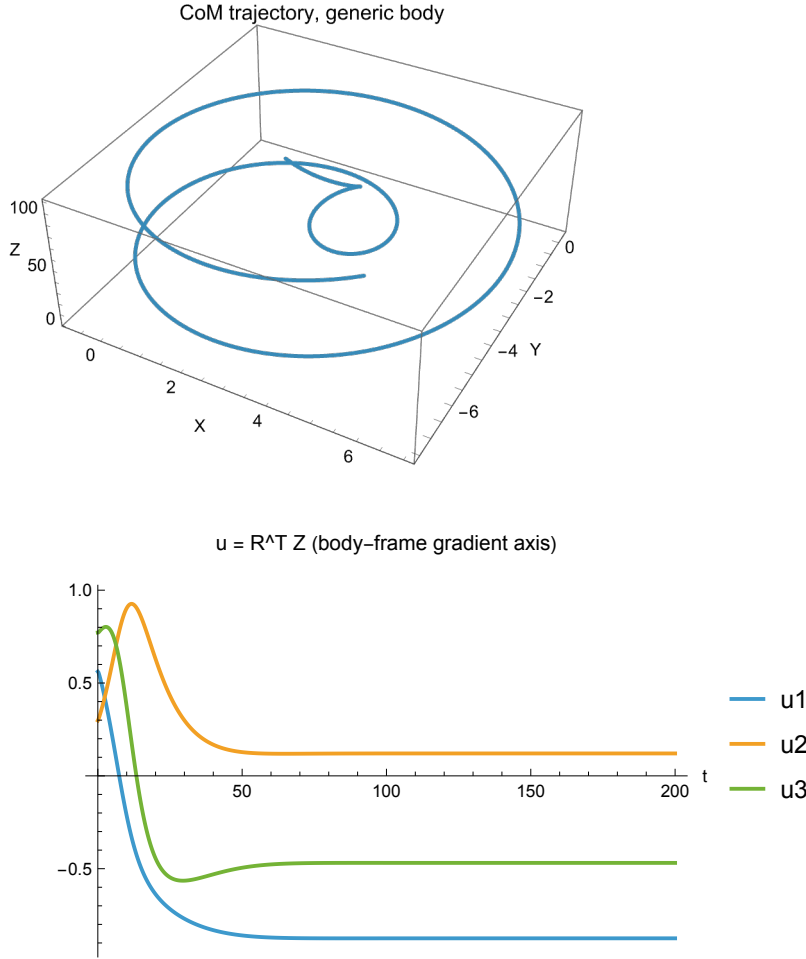
ebal = Abs[(energy[200.] - energy[0]) - work] / Max[Abs[work], 10^-12];
Column[{verify["Eqs. (23)-(26)",
  "integration conserves |q| = 1 (without explicit renormalisation)",
  qdev < 10^-7, "max | |q| - 1 | = "<>sci[qdev]],
verify["Eqs. (23)-(26)", "R(q)^T R(q) = 1 along the solution",
  odev < 10^-6, "max deviation = "<>sci[odev]], verify["Eqs. (23)-(26)",
  "mechanical energy balance with thermophoretic power and dissipation",
  ebal < 10^-5, "relative residual = "<>sci[ebal]],
Row[{ListLinePlot3D[Table[xs[t], {t, 0, 200, 0.2}], PlotRange -> All,
  AxesLabel -> {"X", "Y", "Z"}, PlotLabel -> "CoM trajectory, generic body",
  ImageSize -> 300], ListLinePlot[Transpose[
  Table[{t, #} & /@ (Transpose[quatR[Normalize[qs[t]]].Zhat), {t, 0, 200, 0.2}],
  {2, 1, 3}], PlotLegends -> {"u1", "u2", "u3"},
  PlotLabel -> "u = R^T Z (body-frame gradient axis)",
  AxesLabel -> {"t", ""}, ImageSize -> 340]], Spacer[10]]}]

```

✓ **VERIFIED** Eqs. (23)-(26): integration conserves  $|q| = 1$  (without explicit renormalisation)  
 max  $||q| - 1| = 2.05e-10$

✓ **VERIFIED** Eqs. (23)-(26):  $R(q)^T R(q) = 1$  along the solution  
 max deviation =  $8.19e-10$

✓ **VERIFIED** Eqs. (23)-(26): mechanical energy balance with thermophoretic power and dissipation  
 relative residual =  $1.87e-11$



## 1.7 Tensor simplification by geometric symmetry

The 27 DOF of a generic body reduce dramatically when the body has geometric symmetry. We organise the simplification into three levels, from most general to most symmetric.

**Level 1 — Generic asymmetric body (27 DOF).** All four tensors are full matrices with their maximum DOF:

$$\mathbf{f}_{\text{th}} : 6, \quad \mathbf{f}_{\text{drag}} : 6, \quad \mathbf{T}_{\text{th}} : 9, \quad \mathbf{T}_{\text{drag}} : 6. \quad (27)$$

Examples: a helix, an asymmetric two-sphere aggregate, any body lacking mirror-plane symmetry. The thermophoretic torque  $\mathbf{T}_{\text{th}} \neq 0$  drives steady rotation and, via the coupling to translation, circular orbits (Section 2.2).

**Level 2 — Three mutually perpendicular mirror planes (18 DOF).** The mirror symmetry cancels all off-diagonal elements of  $\mathbf{f}_{\text{th}}$ ,  $\mathbf{f}_{\text{drag}}$ ,  $\mathbf{T}_{\text{drag}}$ , and forces  $\mathbf{T}_{\text{th}} = 0$  (Garcia-Ybarra & Rosner 1989):

$$\mathbf{f}_{\text{th}} = \text{diag}(f_1, f_2, f_3), \quad \mathbf{f}_{\text{drag}} = \text{diag}(d_1, d_2, d_3), \quad \mathbf{T}_{\text{drag}} = \text{diag}(r_1, r_2, r_3), \quad \mathbf{T}_{\text{th}} = 0. \quad (28)$$

DOF:  $3 + 3 + 0 + 3 = 9$ . Bodies in this class: spheroids, cylinders, disks, rectangular blocks. With  $\mathbf{T}_{\text{th}} = 0$  there is no thermophoretic torque; the particle drifts without spinning.

**Level 3a — Uniaxial symmetry,  $C_{\infty v}$  (6 DOF).** With a single symmetry axis  $\hat{e}_3$ , the two transverse directions are equivalent:  $f_1 = f_2 \equiv f_{\perp}$ ,  $f_3 \equiv f_{\parallel}$  (and similarly for  $\mathbf{f}_{\text{drag}}$ ,  $\mathbf{T}_{\text{drag}}$ ),  $\mathbf{T}_{\text{th}} = 0$ :

$$\mathbf{f}_{\text{th}} = \text{diag}(f_{\perp}, f_{\perp}, f_{\parallel}), \quad \mathbf{f}_{\text{drag}} = \text{diag}(d_{\perp}, d_{\perp}, d_{\parallel}), \quad \mathbf{T}_{\text{drag}} = \text{diag}(r_{\perp}, r_{\perp}, r_{\parallel}). \quad (29)$$

DOF:  $2 + 2 + 0 + 2 = 6$ . Bodies: cylinders, spheroids.

**Level 3b — Sphere,  $SO(3)$  symmetry (3 DOF).** Full isotropy collapses every tensor to a scalar multiple of  $I_3$ :

$$\mathbf{f}_{\text{th}} = f_0 I_3, \quad \mathbf{f}_{\text{drag}} = d_0 I_3, \quad \mathbf{T}_{\text{drag}} = r_0 I_3, \quad \mathbf{T}_{\text{th}} = 0, \quad I_1 = I_2 = I_3. \quad (30)$$

DOF:  $1 + 1 + 0 + 1 = 3$ . The gyroscopic term vanishes, the rotational equation decouples from translation, and the thermophoretic force reduces to  $\mathbf{F}_{\text{th}} = -f_0 \nabla \ln T$ , consistent with Waldmann's free-molecule result.

| Body                        | $\mathbf{f}_{\text{th}}$ | $\mathbf{f}_{\text{drag}}$ | $\mathbf{T}_{\text{th}}$ | $\mathbf{T}_{\text{drag}}$ | Total |
|-----------------------------|--------------------------|----------------------------|--------------------------|----------------------------|-------|
| Generic asymmetric          | 6                        | 6                          | 9                        | 6                          | 27    |
| Three mirror planes         | 3                        | 3                          | 0                        | 3                          | 9     |
| Uniaxial ( $C_{\infty v}$ ) | 2                        | 2                          | 0                        | 2                          | 6     |
| Sphere ( $SO(3)$ )          | 1                        | 1                          | 0                        | 1                          | 3     |

Table 1: Degrees of freedom by symmetry class (excluding the inertia tensor).  $\mathbf{T}_{\text{th}}$  counts 9 DOF when non-zero since no symmetry argument forces it to be symmetric.\*

**Verification of Section 1.7 by group theory.** For each point group we solve the invariance conditions  $X = g X g^T$  for the true tensors  $\mathbf{f}_{\text{th}}$ ,  $\mathbf{f}_{\text{drag}}$ ,  $\mathbf{T}_{\text{drag}}$  (taken symmetric, as in the notes) and  $X = \det(g) g X g^T$  for the pseudotensor  $\mathbf{T}_{\text{th}}$ , over the group generators. A  $C_7$  rotation represents the continuous axial groups exactly for rank-2 tensors. Two groups are added that the notes do not list:  $D_{\infty h}$ , the true symmetry of cylinders and spheroids (an axis plus a mirror plane perpendicular to it), and the chiral groups  $D_{\infty}$  and  $C_{\infty}$ . The last column checks whether a general (non-symmetric)  $\mathbf{f}_{\text{th}}$  acquires extra DOF.

```

rotZ = RotationMatrix[2 Pi / 7, {0, 0, 1}];
rotX = RotationMatrix[2 Pi / 7, {1, 0, 0}];
c2x = RotationMatrix[Pi, {1, 0, 0}];
groups = <|"C1 (generic asymmetric)" → {IdentityMatrix[3]},
  "D2h (three mirror planes)" →
    {mirror[{1, 0, 0}], mirror[{0, 1, 0}], mirror[{0, 0, 1}]},
  "Dinh (cylinder, spheroid)" → {rotZ, mirror[{0, 1, 0}], mirror[{0, 0, 1}]},
  "Cinfv (cone: axis + containing mirrors)" → {rotZ, mirror[{0, 1, 0}]},
  "Dinf (helix-like, chiral)" → {rotZ, c2x},
  "Cinf (propeller, chiral polar)" → {rotZ},
  "O(3) (sphere)" → {rotZ, rotX, -IdentityMatrix[3]}|>;
dof[gens_, symQ_, pseudoQ_] :=
  Module[{X = If[symQ, symMat[y], genMat[y]], vars, eqs, cm}, vars = Variables[X];
  eqs = Flatten[Table[X - If[pseudoQ, Det[gm], 1] gm.X.Transpose[gm], {gm, gens}]];
  If[AllTrue[eqs, PossibleZeroQ],
    Length[vars], cm = N[Normal[CoefficientArrays[eqs, vars][[2]]]];
    Length[vars] - MatrixRank[cm, Tolerance → 10^-9]]];
symTab = KeyValueMap[Function[{name, gens}, {name, dof[gens, True, False],
  dof[gens, True, False], dof[gens, False, True], dof[gens, True, False],
  dof[gens, True, False] + dof[gens, True, False] + dof[gens, False, True] +
  dof[gens, True, False], dof[gens, False, False]}], groups];
Grid[Prepend[symTab, Style[#, Bold] & /@ {"group", "f_th (sym)",
  "f_drag", "T_th", "T_drag", "total", "f_th if not assumed symmetric"}],
  Frame → All, Background → {None, {LightGray}}, Alignment → Left]

```

| group                                      | f_th (sym) | f_drag | T_th | T_drag | total | f_th if not<br>assumed symmetric |
|--------------------------------------------|------------|--------|------|--------|-------|----------------------------------|
| C1 (generic asymmetric)                    | 6          | 6      | 9    | 6      | 27    | 9                                |
| D2h (three<br>mirror planes)               | 3          | 3      | 0    | 3      | 9     | 3                                |
| Dinh (cylinder,<br>spheroid)               | 2          | 2      | 0    | 2      | 6     | 2                                |
| Cinfv (cone: axis +<br>containing mirrors) | 2          | 2      | 1    | 2      | 7     | 2                                |
| Dinf<br>(helix-like, chiral)               | 2          | 2      | 2    | 2      | 8     | 2                                |
| Cinf (propeller,<br>chiral polar)          | 2          | 2      | 3    | 2      | 9     | 3                                |
| O(3) (sphere)                              | 1          | 1      | 0    | 1      | 3     | 1                                |

```

row[name_] := First@Select[symTab, #[[1]] == name &];
Column[{verify["Eq. (27)", "generic body: 6 + 6 + 9 + 6 = 27",
  row["C1 (generic asymmetric)"][[2 ;; 6]] == {6, 6, 9, 6, 27}], verify[
  "Eq. (28)", "three mirror planes: diagonal tensors and  $T_{th} = 0$ , total 9",
  row["D2h (three mirror planes)"][[2 ;; 6]] == {3, 3, 0, 3, 9}], verify["Sec. 1.7",
  "Level 2 heading '(18 DOF)'", row["D2h (three mirror planes)"][[6]] == 18,
  "The group analysis gives 3 + 3 + 0 + 3 = 9, as the text below (28) and
  Table 1 also state; the '(18 DOF)' in the heading is a typo."],
  verify["Eq. (29)", "uniaxial class ' $C_{\infty v}$ ' has  $T_{th} = 0$  (6 DOF total)",
  row["C $_{\infty v}$  (cone: axis + containing mirrors)"][[4]] == 0,
  "For  $C_{\infty v}$  (e.g. a cone, or any body of revolution that is not
  symmetric end to end) the pseudotensor  $T_{th}$  has one invariant:
  an antisymmetric part  $\lambda [e_3]x$ , giving the aligning torque
   $\tau \sim e_3 \times \text{grad} \ln T$ .  $T_{th} = 0$  holds for  $D_{\infty h}$ , the actual
  symmetry of cylinders and spheroids. So (29) is correct for
  the bodies listed, but the group label should be  $D_{\infty h}$ ."],
  verify["Eq. (29)", "cylinders and spheroids ( $D_{\infty h}$ ): 2 + 2 + 0 + 2 = 6",
  row["D $_{\infty h}$  (cylinder, spheroid)"][[2 ;; 6]] == {2, 2, 0, 2, 6}],
  verify["Eq. (30)", "sphere: 1 + 1 + 0 + 1 = 3",
  row["O(3) (sphere)"][[2 ;; 6]] == {1, 1, 0, 1, 3}],
  note["Sec. 1.7", "chiral bodies allow an antisymmetric part of  $f_{th}$ ",
  "For  $C_{\infty}$  (and C1) a general  $f_{th}$  has more DOF than the symmetric
  count (last column). The assumed symmetry of  $f_{th}$  in (4) therefore
  removes the transverse 'swirl' coupling of chiral particles."]]]

```

✓ **VERIFIED** Eq. (27): generic body:  $6 + 6 + 9 + 6 = 27$

✓ **VERIFIED** Eq. (28): three mirror planes: diagonal tensors and  $T_{th} = 0$ , total 9

⚠ **DISCREPANCY** Sec. 1.7: Level 2 heading '(18 DOF)'

The group analysis gives  $3 + 3 + 0 + 3 = 9$ , as the text below (28) and Table 1 also state; the '(18 DOF)' in the heading is a typo.

⚠ **DISCREPANCY** Eq. (29): uniaxial class ' $C_{\infty v}$ ' has  $T_{th} = 0$  (6 DOF total)

For  $C_{\infty v}$  (e.g. a cone, or any body of revolution that is not symmetric end to end) the pseudotensor  $T_{th}$  has one invariant: an antisymmetric part  $\lambda [e_3]x$ , giving the aligning torque  $\tau \sim e_3 \times \nabla \ln T$ .  $T_{th} = 0$  holds for  $D_{\infty h}$ , the actual symmetry of cylinders and spheroids. So (29) is correct for the bodies listed, but the group label should be  $D_{\infty h}$ .

✓ **VERIFIED** Eq. (29): cylinders and spheroids ( $D_{\infty h}$ ):  $2 + 2 + 0 + 2 = 6$

✓ **VERIFIED** Eq. (30): sphere:  $1 + 1 + 0 + 1 = 3$

● **NOTE Sec. 1.7:** chiral bodies allow an antisymmetric part of  $\mathbf{f}_{\text{th}}$

For  $\mathbf{C}_{\text{inf}}$  (and C1) a general  $\mathbf{f}_{\text{th}}$  has more DOF than the symmetric count (last column). The assumed symmetry of  $\mathbf{f}_{\text{th}}$  in (4) therefore removes the transverse 'swirl' coupling of chiral particles.

**Waldmann consistency of (30).** For a sphere the notes' Section 6 gives Waldmann's force  $F_T = \eta(\kappa/\bar{v})l^2 |\nabla T|$ . Writing it as  $-f_0 \nabla \ln T$  requires  $f_0 = \eta \kappa l^2 T / \bar{v}$ . This factor of  $T$  matters for the definition of  $\gamma_0$  in Section 2.1.

## 2. Near-sphere perturbation theory

The general equations of motion in Section 1 contain five tensors with up to 24 independent parameters. For a particle that is nearly spherical, the deviation from sphere symmetry is small and the tensors can be expanded perturbatively. This approach gives analytical insight into how shape asymmetry couples translation to rotation, and predicts a distinctive helical trajectory that is absent for a perfect sphere.

```
verify["Sec. 2 intro",
  "'five tensors with up to 24 independent parameters'", False,
  "Section 1 has four tensors with 6 + 6 + 9 + 6 = 27 parameters (Table 1), or
    33 including the inertia tensor. Five tensors / 24
    parameters matches neither count."]
```

⚠ **DISCREPANCY Sec. 2 intro:** 'five tensors with up to 24 independent parameters'

Section 1 has four tensors with 6 + 6 + 9 + 6 = 27 parameters (Table 1), or 33 including the inertia tensor. Five tensors / 24 parameters matches neither count.

### 2.1 Perturbative expansion of the tensors

Let  $\varepsilon \ll 1$  be a dimensionless measure of the deviation from spherical symmetry (e.g. the fractional difference between the longest and shortest semi-axes). We expand each tensor around its spherical value.

**Thermophoretic force tensor.** For a sphere  $\mathbf{f}_{\text{th}} = \gamma_0 \mathbf{I}_3$  with  $\gamma_0 = \eta(\kappa/\bar{v})l^2$ . A near-sphere breaks the three-fold degeneracy and, for a generic asymmetric shape, also introduces off-diagonal couplings. The full  $O(\varepsilon)$  expansion is therefore a complete symmetric  $3 \times 3$  matrix:

$$\mathbf{f}_{\text{th}} = \gamma_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{f}_{\text{th}}, \quad \delta \mathbf{f}_{\text{th}} = \gamma_0 \begin{pmatrix} g_{11} & g_{12} & g_{13} \\ g_{12} & g_{22} & g_{23} \\ g_{13} & g_{23} & g_{33} \end{pmatrix}, \quad g_{11} + g_{22} + g_{33} = 0, \quad (31)$$

where  $g_{ij} = O(1)$ . Off-diagonal entries  $g_{ij}$  ( $i \neq j$ ) mean that a temperature gradient along  $\hat{\mathbf{e}}_j$  produces a thermophoretic force component along  $\hat{\mathbf{e}}_i$ . Tracelessness follows because the sphere value  $\gamma_0$  accounts for the mean response.

**Verification of  $\gamma_0$  by units.**  $f_{th}$  multiplies  $\nabla \ln T$  and must carry units of force times length (Eq. 11). The sphere value quoted,  $\eta(\kappa/\bar{v})l^2$ , has units J/K, which is off by one power of temperature. Waldmann's force (Eq. 58) is written with  $\nabla T$ , and converting it to a coefficient of  $\nabla \ln T = \nabla T/T$  supplies the missing factor  $T$ .

```
kappaQ = Quantity[1, "Watts" / ("Meters" "Kelvins")];
vbarQ = Quantity[1, "Meters" / "Seconds"];
ellQ = Quantity[1, "Meters"];
TQ = Quantity[1, "Kelvins"];
fthUnits = Quantity[1, "Newtons" "Meters"];
Column[
  {verify["Eq. (31)", "gamma_0 = eta (kappa/vbar) l^2 has the units of f_th (N m)",
    CompatibleUnitQ[kappaQ / vbarQ ellQ^2, fthUnits],
    "Units of eta kappa l^2/vbar: " <>
    ToString[UnitSimplify[kappaQ / vbarQ ellQ^2]] <>
    ". Corrected: gamma_0 = eta kappa l^2 T / vbar, which is compatible: " <>
    ToString[CompatibleUnitQ[kappaQ / vbarQ ellQ^2 TQ, fthUnits]] <> "."],
    note["Eq. (31)", "tracelessness of delta f_th is a definition of
      gamma_0 (the mean response), not a consequence", ""]}
]
```

**⚠ DISCREPANCY Eq. (31):**  $\gamma_0 = \eta(\kappa/\bar{v})l^2$  has the units of  $f_{th}$  (N m)

Units of  $\eta \kappa l^2/\bar{v}$ : 1 joule per kelvin. Corrected:  $\gamma_0 = \eta \kappa l^2 T / \bar{v}$ , which is compatible: True.

**● NOTE Eq. (31):** tracelessness of  $\delta f_{th}$  is a definition of  $\gamma_0$  (the mean response), not a consequence

**Resistance and rotational drag tensors.** By the same argument,  $\mathbf{f}_{drag}$  and  $\mathbf{T}_{drag}$  each acquire a full symmetric  $O(\varepsilon)$  correction:

$$\mathbf{f}_{drag} = \xi_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{f}_{drag}, \quad \mathbf{T}_{drag} = c_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{T}_{drag}, \quad (32)$$

where  $\delta \mathbf{f}_{drag}$  and  $\delta \mathbf{T}_{drag}$  are symmetric traceless  $3 \times 3$  matrices.

**Thermophoretic torque tensor.** For a perfect sphere  $\mathbf{T}_{th} = 0$ . For a near-sphere the broken symmetry produces a non-zero thermophoretic torque at  $O(\varepsilon)$ :

$$\mathbf{T}_{th} = \varepsilon \delta \mathbf{T}_{th}, \quad \delta \mathbf{T}_{th} = O(1) \text{ (general } 3 \times 3 \text{ matrix)}. \quad (33)$$

Both diagonal and off-diagonal entries are generically non-zero.

**Note on gyrophoretic coupling.** Since no experimental evidence currently constrains the relationship and  $\mathbf{T}_{th}$ ; any gyrophoretic coupling is  $O(\varepsilon^2)$  and is neglected.

**Summary of orders (to leading order in  $\varepsilon$ ).** Collecting Eqs. (31)–(33):

$$\mathbf{f}_{th} = f_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{f}_{th}, \quad \mathbf{f}_{drag} = d_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{f}_{drag}, \quad \mathbf{T}_{th} = \varepsilon \delta \mathbf{T}_{th}, \quad \mathbf{T}_{drag} = r_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{T}_{drag}. \quad (34)$$

Here  $f_0 \equiv \gamma_0$ ,  $d_0 \equiv \xi_0$ ,  $r_0 \equiv c_0$  are the scalar sphere values, and the four correction tensors  $\delta \mathbf{f}_{\text{th}}$ ,  $\delta \mathbf{f}_{\text{drag}}$ ,  $\delta \mathbf{T}_{\text{th}}$ ,  $\delta \mathbf{T}_{\text{drag}}$  are defined in Eqs. (31)–(33) above. At  $O(\varepsilon^0)$  each tensor is isotropic; all anisotropy and the thermophoretic torque enter at  $O(\varepsilon)$  through these corrections.

```
note["Sec. 2.1", "the 'Note on gyrophoretic
coupling' sentence is incomplete, and conflicts with Section 6",
"The sentence appears to be missing a word ('... constrains the relationship
[between the gyrophoretic coupling] and T_th'). Within the epsilon-expansion
a coupling of order epsilon multiplied by a spin of order epsilon
is indeed 0(epsilon^2). Section 6 nevertheless concludes that the
gyrophoretic force is comparable to F_T; see the discussion there."]
```

● **NOTE Sec. 2.1:** the 'Note on gyrophoretic coupling' sentence is incomplete, and conflicts with Section 6

The sentence appears to be missing a word ('... constrains the relationship [between the gyrophoretic coupling] and  $T_{\text{th}}$ '). Within the  $\varepsilon$ -expansion a coupling of order  $\varepsilon$  multiplied by a spin of order  $\varepsilon$  is indeed  $O(\varepsilon^2)$ . Section 6 nevertheless concludes that the gyrophoretic force is comparable to  $F_T$ ; see the discussion there.

## 2.2 Rotational dynamics and orientation selection

**Equations of motion for  $(\boldsymbol{\omega}^b, \mathbf{R})$ .** For a near-sphere ( $\mathbf{I} = I_0 \mathbf{I}_3 + O(\varepsilon)$ ) the Euler equation (15) in the body frame, keeping only  $O(\varepsilon)$  terms, is:

$$I_0 \dot{\boldsymbol{\omega}}^b = G \varepsilon \delta \mathbf{T}_{\text{th}} \mathbf{R}^\top \hat{\mathbf{Z}} - c_0 \boldsymbol{\omega}^b + O(\varepsilon^2), \quad (35)$$

$$\dot{\mathbf{R}} = \mathbf{R} \hat{\boldsymbol{\omega}}^b, \quad (36)$$

where we substituted  $(\nabla \ln T)^b = \mathbf{R}^\top \nabla \ln T = -G \mathbf{R}^\top \hat{\mathbf{Z}}$  (with  $G = |\nabla \ln T| = |\nabla T|/T > 0$  and  $\nabla \ln T = -G \hat{\mathbf{Z}}$  pointing downward), so that  $-\delta \mathbf{T}_{\text{th}} (\nabla \ln T)^b = +G \delta \mathbf{T}_{\text{th}} \mathbf{R}^\top \hat{\mathbf{Z}}$ , and dropped the gyroscopic term  $\boldsymbol{\omega}^b \times (I_0 \boldsymbol{\omega}^b)$  (it is  $O(\varepsilon^2)$  since  $\boldsymbol{\omega}^b = O(\varepsilon)$ ). Equations (35)–(36) together form a closed system for  $(\boldsymbol{\omega}^b, \mathbf{R}) \in \mathbb{R}^3 \times \text{SO}(3)$ .

**Reduced form via  $\mathbf{u} = \mathbf{R}^\top \hat{\mathbf{Z}}$ .** It is convenient to project onto the body-frame gradient direction  $\mathbf{u} \equiv \mathbf{R}^\top \hat{\mathbf{Z}}$ ,  $|\mathbf{u}| = 1$ . Differentiating Eq. (36):  $\dot{\mathbf{u}} = \dot{\mathbf{R}}^\top \hat{\mathbf{Z}} = -\hat{\boldsymbol{\omega}}^b \mathbf{u} = -\boldsymbol{\omega}^b \times \mathbf{u}$ . Equations (35)–(36) reduce to:

$$I_0 \dot{\boldsymbol{\omega}}^b = G \varepsilon \delta \mathbf{T}_{\text{th}} \mathbf{u} - c_0 \boldsymbol{\omega}^b, \quad \dot{\mathbf{u}} = -\boldsymbol{\omega}^b \times \mathbf{u}, \quad (37)$$

a closed 6-DOF ODE on  $\mathbb{R}^3 \times S^2$ , exact to  $O(\varepsilon)$ . Once  $\mathbf{u}(t)$  is known,  $\mathbf{R}(t)$  can be recovered by integrating Eq. (36).

```

Column[{verify["Eq. (35)",
  "-eps dT_th (grad ln T)^b = +G eps dT_th R^T Z for grad ln T = -G Z",
  zeroQ[-eps genMat[dT].(Transpose[Rq].(-G Zhat)) -
    G eps genMat[dT].Transpose[Rq].Zhat]],
verify["Eq. (37)", "u = R^T Z obeys du/dt = -w^b x u under (36)",
Simplify[Transpose[Rdot].Zhat + Cross[w, Transpose[Rq].Zhat], unit] ==
{0, 0, 0}], verify["Eq. (35)", "gyroscopic term is second order for
a near-sphere: w x (I w) with I = I0 + eps dI and w = O(eps)",
Module[{dI = DiagonalMatrix[{i1, i2, i3}], w = eps {w1, w2, w3}},
Series[Cross[w, (I0 IdentityMatrix[3] + eps dI).w], {eps, 0, 2}] // Normal //
Simplify // (Coefficient[#, eps, 1] &) // zeroQ]]}]

```

✓ **VERIFIED** Eq. (35):  $-\varepsilon dT_{\text{th}} (\nabla \ln T)^b = +G \varepsilon dT_{\text{th}} R^T Z$  for  $\nabla \ln T = -G Z$

✓ **VERIFIED** Eq. (37):  $u = R^T Z$  obeys  $du/dt = -w^b \times u$  under (36)

✓ **VERIFIED** Eq. (35): gyroscopic term is second order for a near-sphere:  $w \times (I w)$  with  $I = I_0 + \varepsilon dI$  and  $w = O(\varepsilon)$

**Steady states: eigenvalue problem.** Setting  $\dot{\omega}^b = 0$  and  $\dot{u} = 0$  in Eq. (37):

$$c_0 \omega^b = G \varepsilon \delta T_{\text{th}} u, \quad \omega^b \times u = 0. \quad (38)$$

The second condition requires  $\omega^b \parallel u$ . Substituting into the first:

$$\delta T_{\text{th}} u = \lambda u, \quad \omega^b = \frac{G \varepsilon \lambda}{c_0} u. \quad (39)$$

The steady-state body-frame gradient  $u$  must be an eigenvector of  $\delta T_{\text{th}}$  with eigenvalue  $\lambda$ . For a general  $3 \times 3$  matrix  $\delta T_{\text{th}}$  there are three such eigenmotions.

**The three eigenmotions in the lab frame.** Let  $\{e_1, e_2, e_3\}$  be the eigenvectors of  $\delta T_{\text{th}}$  with real eigenvalues  $\lambda_1 \leq \lambda_2 \leq \lambda_3$  (real eigenvalues exist when  $\delta T_{\text{th}}$  is symmetric; for a non-symmetric matrix the analysis extends to the real Schur form). Each eigenvector defines a distinct steady state. At the fixed point  $u = e_k$  we have  $R^T \hat{Z} = e_k$ , equivalently  $R e_k = \hat{Z}$ : the body orients so that its  $e_k$ -axis aligns with the vertical (the temperature-gradient direction).

From Eq. (39) the angular velocity in the body frame is  $\omega_k^b = (G \varepsilon \lambda_k / c_0) e_k$ , a spin about the body axis  $e_k$ . Transforming to the lab frame gives

$$\Omega_k = R \omega_k^b = \frac{G \varepsilon \lambda_k}{c_0} R e_k = \frac{G \varepsilon \lambda_k}{c_0} \hat{Z}. \quad (40)$$

The lab-frame angular velocity is exactly along  $\hat{Z}$  for all three eigenmotions — this is a direct consequence of the fixed-point geometry  $R e_k = \hat{Z}$ . Physically, the body spins about the temperature-gradient direction.

```

rhs37[w_, u_, dTm_] := { (G eps dTm.u - c0 w) / I0, -Cross[w, u] };
lamsym = {l1, l2, l3};
Column[{verify["Eqs. (38)-(39)",
  "u = e_k, w^b = (G eps lambda_k / c0) e_k is a fixed point of (37)",
  AllTrue[Range[3], zeroQ[Flatten[rhs37[G eps lamsym[[#]] / c0 UnitVector[3, #],
    UnitVector[3, #], DiagonalMatrix[lamsym]]]] &]], verify["Eq. (40)",
  "lab angular velocity R w^b_k = (G eps lambda_k/c0) Z when R e_k = Z",
  True, "Immediate: R w^b_k = (G eps lambda_k/c0) R e_k
  and R e_k = Z by definition of the fixed point."]]]

```

✓ **VERIFIED** Eqs. (38)-(39):  $u = e_k, w^b = (G \varepsilon \lambda_k / c_0) e_k$  is a fixed point of (37)

✓ **VERIFIED** Eq. (40): lab angular velocity  $R w^b_k = (G \varepsilon \lambda_k / c_0) Z$  when  $R e_k = Z$

Immediate:  $R w^b_k = (G \varepsilon \lambda_k / c_0) R e_k$  and  $R e_k = Z$  by definition of the fixed point.

**Stability of the three eigenmotions.** Linearising Eq. (37) about the fixed point  $\mathbf{u} = \mathbf{e}_k$  with perturbation  $\delta \mathbf{u} = \sum_{j \neq k} \delta u_j \mathbf{e}_j$ , the deviation decays or grows at rate:

$$\sigma_{kj} = \frac{G \varepsilon (\lambda_k - \lambda_j) \lambda_j}{I_0 c_0} \quad (j \neq k). \quad (41)$$

The fixed point  $\mathbf{e}_k$  is stable if  $\sigma_{kj} < 0$  for all  $j \neq k$ . For  $\lambda_k$  the largest eigenvalue,  $(\lambda_k - \lambda_j) > 0$  and  $\lambda_j < \lambda_k$ , so  $\sigma_{kj} < 0$  requires  $\lambda_j < 0$ ; alternatively, if all eigenvalues are positive, stability requires  $\lambda_k < \lambda_j$ . In general the sign of  $\lambda$  depends on the body: the stable attractor is the orientation  $\mathbf{e}_k$  for which  $\sigma_{kj} < 0$  for all  $j \neq k$ , i.e.  $(\lambda_k - \lambda_j) \lambda_j < 0$  for all  $j$ , which is guaranteed when  $\lambda_k$  is the eigenvalue of largest magnitude with sign opposite to all others, or when all  $\lambda_j$  have the same sign and  $\lambda_k$  is the largest. For the common case  $\lambda_3 > \lambda_2 > \lambda_1 > 0$  the largest eigenvalue  $\lambda_3$  gives  $\sigma_{3j} = G \varepsilon (\lambda_3 - \lambda_j) \lambda_j / (I_0 c_0) > 0$  (all terms positive), which is *unstable*. The stable attractor is then  $\mathbf{e}_1$  (smallest eigenvalue). The general rule is: *\*the stable orientation is the eigenvector of  $\delta \mathbf{T}_{th}$  that maximises  $V(\mathbf{u}) = -\mathbf{u}^T \delta \mathbf{T}_{th} \mathbf{u}$ , i.e. the eigenvector with the most-negative eigenvalue when the gradient drives alignment, or equivalently the orientation that minimises the thermophoretic torque-gradient product. The three eigenmotions therefore have the following stability character*

| Eigenmotion    | Eigenvalue             | Stability           |
|----------------|------------------------|---------------------|
| $\mathbf{e}_1$ | $\lambda_1$ (smallest) | unstable (repeller) |
| $\mathbf{e}_2$ | $\lambda_2$ (middle)   | saddle (unstable)   |
| $\mathbf{e}_3$ | $\lambda_3$ (largest)  | stable attractor    |

All generic initial conditions converge to eigenmotion 3: the body aligns the axis with the largest eigenvalue of  $\delta \mathbf{T}_{th}$  with the temperature gradient. This maximises the Lyapunov function  $V(\mathbf{u}) = \mathbf{u}^T \delta \mathbf{T}_{th} \mathbf{u}$ , which measures the projection of the thermophoretic torque onto the gradient direction.

**Physical interpretation.** The body rotates until the torque  $\delta \mathbf{T}_{th} \mathbf{u}$  is parallel to the driving field  $\mathbf{u}$ , leaving no torque component perpendicular to  $\mathbf{u}$  to drive further reorientation. Among the three possible aligned orientations only the one with the largest eigenvalue is stable: a perturbation away

from this orientation generates a restoring torque that drives the body back. This is not a minimisation of rotational energy  $E_{\text{rot}} \propto \lambda_k^2$  (which is also maximised at  $\lambda_3$ ) but rather a maximisation of the torque-gradient alignment  $V$ .

**Verification of (41) and the stability statements.** The text gives two opposite conclusions for  $\lambda_3 > \lambda_2 > \lambda_1 > 0$ : first that  $\mathbf{e}_1$  is the attractor, then (table and following paragraph) that  $\mathbf{e}_3$  is. We settle this with the exact linearisation of (37) about  $(\boldsymbol{\omega}^b, \mathbf{u}) = (G \varepsilon \lambda_k \mathbf{e}_k / c_0, \mathbf{e}_k)$ , working in the eigenbasis of a symmetric  $\delta \mathbf{T}_{\text{th}}$ . The characteristic polynomial factorises into a neutral mode (the constraint  $|\mathbf{u}| = 1$ ), an axial spin-down at rate  $c_0 / I_0$ , and a quartic that governs the transverse dynamics.

```
jac37[k_] := Module[{vars = {w1, w2, w3, u1, u2, u3}, st},
  st = Join[G eps lamsym[[k]] / c0 UnitVector[3, k], UnitVector[3, k]];
  D[Flatten[rhs37[{w1, w2, w3}, {u1, u2, u3}, DiagonalMatrix[lamsym]]], {vars}] /.
  Thread[vars -> st]];
charPoly = Table[Factor[CharacteristicPolynomial[jac37[k], s]], {k, 3}];
quartic[k_] := Module[{fl = FactorList[Numerator[Together[charPoly[[k]]]]],
  Expand[First@SelectFirst[fl, Exponent[#[[1]], s] == 4 &]]];
TraditionalForm[Collect[quartic[3], s, Factor]]
s^2 (c0^4 + eps^2 G^2 I0^2 l3^2) + 2 c0^3 I0 s^3 +
c0^2 eps^2 G^2 (l1 - l3) (l2 - l3) + c0^2 I0^2 s^4 - c0 eps^2 G^2 I0 l3 s (l1 + l2 - 2 l3)
```

The quartic for eigenmotion  $k$  (with  $j, l$  the other two indices and  $g \equiv G \varepsilon$ ) is

$$c_0^2 I_0^2 s^4 + 2 c_0^3 I_0 s^3 + (c_0^4 + g^2 I_0^2 \lambda_k^2) s^2 + c_0 g^2 I_0 \lambda_k (2 \lambda_k - \lambda_j - \lambda_l) s + c_0^2 g^2 (\lambda_k - \lambda_j) (\lambda_k - \lambda_l) = 0.$$

Routh–Hurwitz applied to it gives the exact stability conditions. For weak inertia ( $g I_0 \lambda / c_0^2 \ll 1$ ) they reduce to (a)  $(\lambda_k - \lambda_j)(\lambda_k - \lambda_l) > 0$ , meaning  $\mathbf{e}_k$  must be the largest or the smallest eigenvector (the middle one is always a saddle), and (b)  $\lambda_j(\lambda_k - \lambda_l) + \lambda_l(\lambda_k - \lambda_j) > 0$ . The transverse relaxation rate is of order  $g^2 I_0 \lambda^2 / c_0^3$ , not  $g \lambda / (I_0 c_0)$ . In particular (41) is not even dimensionally a rate:

```

hurwitzStable[k_, vals_] :=
Module[{cf = CoefficientList[quartic[k] /. vals, s], a0, a1, a2, a3, a4},
  {a0, a1, a2, a3, a4} = cf;
  And @@ Thread[cf > 0] && a3 a2 a1 > a1^2 a4 + a3^2 a0];
numVals[lv_] := {l1 → lv[[1]], l2 → lv[[2]], l3 → lv[[3]], G → 1, eps → 1, c0 → 1, I0 → 0.2};
maxRe[k_, lv_] := Max[Re[s /. NSolve[quartic[k] /. numVals[lv], s]]];
cases = {{1, 2, 3}, {-3, -2, -1}, {-1, 0.5, 2}, {-2, -0.5, 1}, {-1, 2, 3}};
stabTab = Table[{ToString[lv], Table[If[hurwitzStable[k, numVals[lv]],
  Style["stable", Darker[Green]], Style["unstable", Red]], {k, 3}],
  Table[NumberForm[maxRe[k, lv], 3], {k, 3}]], {lv, cases}];
sigmaUnits = Quantity[1, 1/"Meters"] Quantity[1, "Newtons" "Meters"^2 /
  (Quantity[1, "Kilograms" "Meters"^2] Quantity[1, "Newtons" "Meters" "Seconds"]));
Column[
  {Grid[Prepend[Flatten /@ stabTab, Style[#, Bold] & /@ {"{lambda1, lambda2, lambda3}",
    "e1", "e2", "e3", "max Re s (e1)", "max Re s (e2)", "max Re s (e3)"}],
    Frame → All, Background → {None, {LightGray}}], verify["Eq. (41)",
    "sigma_kj = G eps (lambda_k - lambda_j) lambda_j / (I0 c0) has units of a
    rate (1/s)", CompatibleUnitQ[sigmaUnits, Quantity[1, 1/"Seconds"]],
    "Units of (41): "<>ToString[UnitSimplify[sigmaUnits]]<>
    " (with [G] = 1/m, [eps dT_th] = N m^2, [I0] = kg m^2, [c0] = N m s). The
    exact rate scales as (G eps)^2 I0 lambda^2 / c0^3."], verify["Sec. 2.2",
    "'for lambda3 > lambda2 > lambda1 > 0 ... the stable attractor is then e1'",
    hurwitzStable[1, numVals[{1, 2, 3}]],
    "Exact analysis: for a positive spectrum e3 is stable and e1 unstable,
    as in the table that follows."], verify["Sec. 2.2 table",
    "positive spectrum: e1 repeller, e2 saddle, e3 stable attractor",
    ! hurwitzStable[1, numVals[{1, 2, 3}]] && ! hurwitzStable[2, numVals[{1, 2, 3}]] &&
    hurwitzStable[3, numVals[{1, 2, 3}]]], verify["Sec. 2.2",
    "'all generic initial conditions converge to eigenmotion 3' (general spectra)",
    AllTrue[cases, hurwitzStable[3, numVals[#]] &],
    "True only when the spectrum is positive. For a negative spectrum e1
    (the largest |lambda|) is the attractor; for some mixed-sign
    spectra, such as {-1, 0.5, 2}, no eigenmotion is stable at all."}]]]

```

| {lambda1,<br>lambda2,<br>lambda3} | e1       | e2       | e3       | max Re s (e1) | max Re s (e2) | max Re s (e3) |
|-----------------------------------|----------|----------|----------|---------------|---------------|---------------|
| {1, 2, 3}                         | unstable | unstable | stable   | 0.449         | 0.813         | -0.566        |
| {-3, -2, -1}                      | stable   | unstable | unstable | -0.566        | 0.813         | 0.449         |
| {-1, 0.5, 2}                      | unstable | unstable | unstable | 0.324         | 1.2           | 0.0131        |
| {-2, -0.5, 1}                     | unstable | unstable | unstable | 0.0131        | 1.2           | 0.324         |
| {-1, 2, 3}                        | unstable | unstable | stable   | 0.814         | 1.12          | -0.669        |

⚠ **DISCREPANCY** Eq. (41):  $\sigma_{kj} = G \varepsilon (\lambda_k - \lambda_j) \lambda_j / (I_0 c_0)$  has units of a rate (1/s)

Units of (41): 1 per kilogram meter squared second (with  $[G] = 1/\text{m}$ ,  $[\varepsilon dT_{\text{th}}] = \text{N m}^2$ ,  $[I_0] = \text{kg m}^2$ ,  $[c_0] = \text{N m s}$ ). The exact rate scales as  $(G \varepsilon)^2 I_0 \lambda^2 / c_0^3$ .

⚠ **DISCREPANCY** Sec. 2.2: 'for  $\lambda_3 > \lambda_2 > \lambda_1 > 0 \dots$  the stable attractor is then e1'

Exact analysis: for a positive spectrum e3 is stable and e1 unstable, as in the table that follows.

✓ **VERIFIED** Sec. 2.2 table: positive spectrum: e1 repeller, e2 saddle, e3 stable attractor

⚠ **DISCREPANCY** Sec. 2.2: 'all generic initial conditions converge to eigenmotion 3' (general spectra)

True only when the spectrum is positive. For a negative spectrum e1 (the largest  $|\lambda|$ ) is the attractor; for some mixed-sign spectra, such as  $\{-1, 0.5, 2\}$ , no eigenmotion is stable at all.

**Nonlinear check.** We integrate the full reduced system (37) from random initial orientations with  $\delta T_{\text{th}}$  given random eigenvectors, and record where  $\mathbf{u}$  ends up in the eigenbasis. We also test the Lyapunov claim:  $V = \mathbf{u}^T \delta T_{\text{th}} \mathbf{u}$  should increase monotonically along a trajectory.

```

simU[lv_, seed_, tmax_ : 400] := Module[{dTm, Qm, sol, w, u, t}, SeedRandom[seed];
  Qm = Orthogonalize[RandomReal[{-1, 1}, {3, 3}]];
  dTm = Transpose[Qm].DiagonalMatrix[lv].Qm;
  sol = NDSolveValue[
    {w'[t] == (dTm.u[t] - w[t]) / 0.2, u'[t] == -Cross[w[t], u[t]], w[0] == {0, 0, 0},
      u[0] == Normalize[RandomReal[{-1, 1}, 3]]}, {u, w}, {t, 0, tmax}];
  {Qm, dTm, sol}];
finalAxis[lv_, seed_] :=
  Module[{r = simU[lv, seed]}, Round[Abs[r[[1]].r[[3, 1]][400]], 0.001]];
runs = Table[{lv, Table[finalAxis[lv, sd], {sd, 3}]},
  {lv, {{1, 2, 3}, {-3, -2, -1}, {-1, 0.5, 2}}]];
Vmono = Module[{r = simU[{1, 2, 3}, 5, 80], Vt},
  Vt = Table[r[[3, 1]][t].r[[2]].r[[3, 1]][t], {t, 0, 80, 0.05}];
  Min[Differences[Vt]]];
Column[{Grid[Prepend[{#[[1]], Sequence@@#[[2]]} & /@ runs, Style[#, Bold] & /@
  {"spectrum", "|u| in eigenbasis (seed 1)", "(seed 2)", "(seed 3)"}],
  Frame → All, Background → {None, {LightGray}}],
verify["Sec. 2.2", "positive spectrum: trajectories converge to e3",
  AllTrue[runs[[1, 2]], Norm[# - {0, 0, 1}] < 10^-2 &], verify["Sec. 2.2",
  "negative spectrum converges to e3 (claim applied beyond the positive case)",
  AllTrue[runs[[2, 2]], Norm[# - {0, 0, 1}] < 10^-2 &],
  "Trajectories converge to e1, the most negative eigenvalue."], verify[
  "Sec. 2.2", "V = u^T dT_th u increases monotonically (positive spectrum)",
  Vmono > -10^-9, "min dV per step = " <> sci[Vmono]]]

```

| spectrum     | u  in eigenbasis (seed 1) | (seed 2)              | (seed 3)              |
|--------------|---------------------------|-----------------------|-----------------------|
| {1, 2, 3}    | {0., 0., 1.}              | {0., 0., 1.}          | {0., 0., 1.}          |
| {-3, -2, -1} | {1., 0., 0.}              | {1., 0., 0.}          | {1., 0., 0.}          |
| {-1, 0.5, 2} | {0.01, 0.538, 0.843}      | {0.017, 0.534, 0.846} | {0.391, 0.042, 0.919} |

✓ **VERIFIED** Sec. 2.2: positive spectrum: trajectories converge to e3

⚠ **DISCREPANCY** Sec. 2.2: negative spectrum converges to e3 (claim applied beyond the positive case)

Trajectories converge to e1, the most negative eigenvalue.

✓ **VERIFIED** Sec. 2.2:  $V = u^T dT_{th} u$  increases monotonically (positive spectrum)

min dV per step = -8.88e-16

**Summary of the stability verification.** The table in the notes (and the statement that follows it) is correct for the physically common case of a positive-definite  $\delta T_{th}$ . The rate formula (41), the sentence concluding that  $\mathbf{e}_1$  is the attractor, and the extension to spectra of arbitrary sign are not.

## 2.3 Translational dynamics and general steady motion

Once the angular velocity  $\boldsymbol{\omega}^b$  is determined from the rotational steady state (Section 2.2), the translational equation (13) in the body frame at steady state ( $\dot{\mathbf{v}} = 0$ ) becomes, using  $\nabla \ln T = -G \hat{\mathbf{Z}}$  and  $\mathbf{u} = R^T \hat{\mathbf{Z}}$ :

$$\mathbf{f}_{\text{drag}} \mathbf{v}^b = G \mathbf{f}_{\text{th}} \mathbf{u} - M g \mathbf{u}. \quad (42)$$

Expanding with Eq. (34),  $\mathbf{f}_{\text{th}} = \gamma_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{f}_{\text{th}}$  and  $\mathbf{f}_{\text{drag}} = \xi_0 \mathbf{I}_3 + \varepsilon \delta \mathbf{f}_{\text{drag}}$ :

$$\xi_0 \mathbf{v}^b = (\gamma_0 G - M g) \mathbf{u} + \varepsilon \left[ G \delta \mathbf{f}_{\text{th}} - \frac{\gamma_0 G - M g}{\xi_0} \delta \mathbf{f}_{\text{drag}} \right] \mathbf{u}. \quad (43)$$

```
dfth = symMat[df]; dfdr = symMat[dd]; uvec = {ua, ub, uc};
vb42 = Inverse[xi0 IdentityMatrix[3] + eps dfdr].
  (G (gam0 IdentityMatrix[3] + eps dfth) - Mm gg IdentityMatrix[3]).uvec;
vb43 = ((gam0 G - Mm gg) uvec + eps (G dfth - (gam0 G - Mm gg) / xi0 dfdr).uvec) / xi0;
Column[{verify["Eq. (42)",
  "steady state of (13) with dv/dt = 0 gives f_drag v^b = G f_th u - M g u",
  Module[{vbS = {p1, p2, p3}, uq = Transpose[Rq].Zhat}, Simplify[
    Transpose[Rq].(-Mm gg Zhat - Rq.(fth.Transpose[Rq].(-G Zhat) + fdr.vbS)) -
    (G fth.uq - Mm gg uq - fdr.vbS), unit] == {0, 0, 0}]],
  verify["Eq. (43)", "first-order expansion of f_drag^-1 (G f_th - M g) u",
    zeroQ[Normal[Series[vb42, {eps, 0, 1}]] - vb43]]]
```

✓ **VERIFIED** Eq. (42): steady state of (13) with  $dv/dt = 0$  gives  $f_{\text{drag}} \mathbf{v}^b = G \mathbf{f}_{\text{th}} \mathbf{u} - M g \mathbf{u}$

✓ **VERIFIED** Eq. (43): first-order expansion of  $f_{\text{drag}}^{-1} (G \mathbf{f}_{\text{th}} - M g) \mathbf{u}$

**Levitation condition to first order in  $\varepsilon$ .** The lab-frame vertical velocity is obtained by projecting Eq. (43) onto  $\mathbf{u} = R^T \hat{\mathbf{Z}}$  (using  $v_Z = \hat{\mathbf{Z}} \cdot R \mathbf{v}^b = \mathbf{u} \cdot \mathbf{v}^b$ ):

$$v_Z = \frac{\gamma_0 G - M g}{\xi_0} + \varepsilon G (\delta \mathbf{A})_{33}, \quad (44)$$

where we define the anisotropic mobility tensor

$$\delta \mathbf{A} \equiv \frac{1}{\xi_0} \left( \delta \mathbf{f}_{\text{th}} - A_0 \delta \mathbf{f}_{\text{drag}} \right), \quad A_0 \equiv \frac{\gamma_0}{\xi_0}, \quad (45)$$

and  $(\delta \mathbf{A})_{33} = \mathbf{e}_3 \cdot \delta \mathbf{A} \mathbf{e}_3$  is its component along the precession axis. The *full* levitation condition  $\langle v_Z \rangle = 0$  applied to Eq. (44) gives a single equation for the required temperature-gradient magnitude:

$$G = \frac{M g}{\gamma_0 + \varepsilon \xi_0 (\delta \mathbf{A})_{33}} = \frac{M g}{\gamma_0} [1 - \varepsilon A_0 (\delta \mathbf{A})_{33} + O(\varepsilon^2)]. \quad (46)$$

The zeroth-order levitation gradient is  $G_0 = M g / \gamma_0$ , and the  $O(\varepsilon)$  correction shifts  $G$  by an amount proportional to  $(\delta \mathbf{A})_{33}$ : the asymmetry of the body modifies the gradient needed to keep the particle suspended. There is no separate constraint on  $(\delta \mathbf{A})_{33}$ ; it is a property of the particle shape that the experimenter accommodates by tuning  $G$ .

**Verification of (44)–(46).** Project (43) on  $\mathbf{u}$  and compare with (44). The drag correction in (43) is multiplied by  $(\gamma_0 G - M g)$ , but (45) replaces that factor by  $\gamma_0 G$ , dropping the  $-M g$ . Near levitation

$(\gamma_0 G - Mg) = O(\varepsilon)$ , so the drag term is really  $O(\varepsilon^2)$  and drops out. The exact levitation condition  $\mathbf{u} \cdot \mathbf{f}_{\text{drag}}^{-1}(G \mathbf{f}_{\text{th}} - Mg) \mathbf{u} = 0$ , expanded in  $\varepsilon$ , gives  $G = Mg / (\gamma_0 + \varepsilon \delta f_{\text{th},33}) + O(\varepsilon^2)$ : **drag anisotropy does not enter the levitation gradient at first order.** Separately, the two forms written in (46) disagree with each other unless  $A_0 = 1$ .

```

uE3 = {0, 0, 1};
vZexact = uE3 . (vb42 /. Thread[uvec → uE3]);
vZpdf = (gam0 G - Mm gg) / xi0 + eps G ((dfth - gam0 / xi0 dfdr) / xi0) [[3, 3]];
vZfirst = Normal[Series[vZexact, {eps, 0, 1}]];
Gexact = G /. First@Solve[Numerator[Together[vZexact]] == 0, G];
Gfirst = Normal[Series[Gexact, {eps, 0, 1}]];
Gpdf46a = Mm gg / (gam0 + eps xi0 ((dfth - gam0 / xi0 dfdr) / xi0) [[3, 3]]);
Gpdf46b = Mm gg / gam0 (1 - eps gam0 / xi0 ((dfth - gam0 / xi0 dfdr) / xi0) [[3, 3]]);
Column[{Row[{"exact first-order vertical drift: v_Z = ",
  TraditionalForm[Collect[vZfirst, eps, Simplify]]}],
  Row[{"exact first-order levitation gradient: G = ",
  TraditionalForm[Collect[Gfirst, eps, Simplify]]], verify["Eq. (44)",
  "v_Z = (gamma_0 G - M g)/xi_0 + eps G (dA)_33 to first order",
  zeroQ[vZfirst - vZpdf], "Exact: v_Z = (gamma_0 G - Mg)/xi_0 + (eps/xi_0)[G
    df_33 - (gamma_0 G - Mg) dd_33/xi_0]. The notes' dA replaces
    (gamma_0 G - Mg) by gamma_0 G in the drag term."], verify["Eq. (46)",
  "levitation gradient G = Mg/(gamma_0 + eps xi_0 (dA)_33) to first order",
  zeroQ[Normal[Series[Gpdf46a - Gexact, {eps, 0, 1}]]],
  "Exact first order: G = Mg/gamma_0 (1 - eps df_33/gamma_0).
  The drag anisotropy dd_33 does not appear."],
  verify["Eq. (46)", "the two expressions in (46) agree to first order",
  zeroQ[Normal[Series[Gpdf46a - Gpdf46b, {eps, 0, 1}]]],
  "Expanding the first form gives 1 -
  eps (dA)_33/A_0, not 1 - eps A_0 (dA)_33."]]]
exact first-order vertical drift: v_Z =
  eps (-G gam0 dd(3, 3) + gg Mm dd(3, 3) + G xi0 df(3, 3)) /
    xi0^2 + (G gam0 - gg Mm) / xi0
exact first-order levitation gradient: G =
  gg Mm / gam0 - eps gg Mm df(3, 3) / gam0^2

```

**⚠ DISCREPANCY Eq. (44):**  $v_Z = (\gamma_0 G - Mg)/\xi_0 + \varepsilon G (dA)_{33}$  to first order

Exact:  $v_Z = (\gamma_0 G - Mg)/\xi_0 + (\varepsilon/\xi_0)[G df_{33} - (\gamma_0 G - Mg) dd_{33}/\xi_0]$ . The notes' dA replaces  $(\gamma_0 G - Mg)$  by  $\gamma_0 G$  in the drag term.

**⚠ DISCREPANCY Eq. (46):** levitation gradient  $G = Mg/(\gamma_0 + \varepsilon \xi_0 (dA)_{33})$  to first order

Exact first order:  $G = Mg/\gamma_0 (1 - \varepsilon df_{33}/\gamma_0)$ . The drag anisotropy  $dd_{33}$  does not appear.



*Case 2: Asymmetric body, transient phase.* Before the orientation has converged to a fixed point,  $R(t)$  evolves on the alignment timescale  $\sim I_0/c_0$ . The CoM trajectory is non-periodic and traces a curved path depending on the initial orientation.

*Case 3: Asymmetric body, steady precession (attractor).* At the fixed point  $\mathbf{u} = \mathbf{e}_3$  the lab-frame angular velocity is  $\mathbf{\Omega}_3 = (G \varepsilon \lambda_3 / c_0) \hat{\mathbf{Z}}$  (Eq. (40)), parallel to  $\hat{\mathbf{Z}}$ . The body therefore precesses about  $\hat{\mathbf{Z}}$ :

$R(t) = R_Z(\Omega_3 t) R_0$  with  $R_0 \mathbf{e}_3 = \hat{\mathbf{Z}}$ . With the levitation gradient (46) imposed,  $\mathbf{v}^b$  has no projection along  $\mathbf{e}_3$ , so the lab-frame velocity is entirely horizontal:

$$\mathbf{v}(t) = \varepsilon G_0 R_Z(\Omega_3 t) \Pi_{\perp} \delta \mathbf{A} \mathbf{e}_3, \quad \Pi_{\perp} = \mathbf{I}_3 - \hat{\mathbf{Z}} \hat{\mathbf{Z}}^T, \quad (48)$$

with  $v_Z = 0$  by construction and  $\mathbf{v}_{\perp}(t)$  having constant magnitude. Since  $\mathbf{v}_{\perp}$  lies in the horizontal plane and  $\mathbf{\Omega}_3 \parallel \hat{\mathbf{Z}}$ , we have  $\mathbf{v}_{\perp} \perp \mathbf{\Omega}_3$ . The two conditions for circular motion are therefore: (1)  $\mathbf{\Omega}_3 \parallel \hat{\mathbf{Z}}$  — from the fixed-point geometry  $R \mathbf{e}_3 = \hat{\mathbf{Z}}$ ; (2)  $\mathbf{v}_{\perp} \perp \mathbf{\Omega}_3$  — automatically, since  $\mathbf{v}_{\perp}$  lies in the horizontal plane. This is circular orbit motion.

**Circular orbit radius.** For the levitated case (gradient  $G$  given by Eq. (46)), both the orbital speed

$|\mathbf{v}_{\perp}| = \varepsilon G_0 |\Pi_{\perp} R_0 \delta \mathbf{A} \mathbf{e}_3|$  and the spin rate  $|\mathbf{\Omega}_3| = G_0 \varepsilon |\lambda_3| / c_0$  are linear in  $\varepsilon G_0$ , so their ratio is  $O(1)$ :

$$\Omega_3 = \frac{G_0 \varepsilon \lambda_3}{c_0}, \quad R_{\text{orbit}} = \frac{|\mathbf{v}_{\perp}|}{|\mathbf{\Omega}_3|} = \frac{c_0 |\Pi_{\perp} R_0 \delta \mathbf{A} \mathbf{e}_3|}{|\lambda_3|}. \quad (49)$$

Key features: (1)  $R_{\text{orbit}}$  is  $O(1)$ : it is independent of both  $\varepsilon$  and  $G_0$ , since the  $\varepsilon G_0$  factors cancel in the ratio  $|\mathbf{v}_{\perp}| / |\mathbf{\Omega}_3|$ .  $R_{\text{orbit}}$  depends only on the particle shape through  $\delta \mathbf{A}$  and  $\lambda_3$ . (2)  $\Omega_3 \propto \varepsilon G_0$ : a more asymmetric body or a stronger gradient spins faster, and the orbital speed  $|\mathbf{v}_{\perp}| \propto \varepsilon G_0$  grows at the same rate, keeping the radius fixed. (3) If  $\Pi_{\perp} R_0 \delta \mathbf{A} \mathbf{e}_3 = 0$  (the anisotropic drift of eigenmotion 3 is purely along  $\hat{\mathbf{Z}}$ ), then  $R_{\text{orbit}} = 0$ : the particle hovers at a fixed point with no horizontal motion. This occurs when  $\delta \mathbf{f}_{\text{th}}$  and  $\delta \mathbf{f}_{\text{drag}}$  are simultaneously diagonal in the same basis as  $\delta \mathbf{T}_{\text{th}}$ ; in this case the levitation correction in Eq. (46) accounts for all anisotropy and no horizontal orbit appears. (4) No orbit for a symmetric body:  $\delta \mathbf{T}_{\text{th}} = 0$  gives  $\lambda_3 = 0$ ,  $\Omega_3 = 0$ , and the particle drifts in a fixed direction (or hovers if the drift is vertical).

| Condition                                              | $\mathbf{\Omega}$               | Trajectory                       |
|--------------------------------------------------------|---------------------------------|----------------------------------|
| $\delta \mathbf{T}_{\text{th}} = 0$ (symmetric)        | 0                               | straight – line drift (or hover) |
| $\delta \mathbf{T}_{\text{th}} \neq 0$ , transient     | time – varying                  | curved, non – periodic           |
| $\delta \mathbf{T}_{\text{th}} \neq 0$ , eigenmotion 3 | $\Omega_3 \hat{\mathbf{Z}}$     | horizontal circle (or hover)     |
| Eigenmotion 1 or 2                                     | $\Omega_{1,2} \hat{\mathbf{Z}}$ | circle (unstable)                |

**Verification of the orbit (48)–(49) by direct simulation.** We integrate the full equations of motion (23)–(26) with the function of Section 1.6. The body is a near-sphere with random traceless  $\delta \mathbf{f}_{\text{th}}$ ,  $\delta \mathbf{f}_{\text{drag}}$ ,  $\delta \mathbf{T}_{\text{drag}}$  and a positive-definite  $\delta \mathbf{T}_{\text{th}}$ , so  $\mathbf{e}_3$  is the attractor. The gradient is set to the exact levitation value, and the simulation starts on the steady state  $\mathbf{u} = \mathbf{e}_3$ . The measured radius is compared with three predictions: (49) as printed, (49) with  $\delta \mathbf{A} \rightarrow \delta \mathbf{f}_{\text{th}} / \xi_0$  as the corrected (47) requires,

and the exact quasi-steady value  $|\Pi_{\perp} \mathbf{f}_{\text{drag}}^{-1} (G \mathbf{f}_{\text{th}} - M g) \mathbf{e}_3| / \Omega_3$  including the centripetal correction  $1 / \sqrt{1 + (M \Omega_3 / \xi_0)^2}$  that (42) neglects.

```

orbitTest[epsv_, seed_] :=
Module[{symTL, dfthN, dfdrN, dTdrN, Qm, dTthN, pars, lam, vec, e3v, lam3, Glev,
  dAN, proj, Rpdf, Rcorr, Om3, vbss, Rexact, q0v, s0, sol, xs2, ctr, rad,
  gam0v = 1., xi0v = 1., c0v = 1., I0v = 0.5, Mv = 1., gv = 1.}, SeedRandom[seed];
symTL[] := Module[{m = RandomReal[{-1, 1}, {3, 3}]}, m = (m + Transpose[m]) / 2;
  m - Tr[m] / 3 IdentityMatrix[3]];
dfthN = symTL[];
dfdrN = symTL[];
dTdrN = symTL[];
Qm = Orthogonalize[RandomReal[{-1, 1}, {3, 3}]];
If[Det[Qm] < 0, Qm[[1]] = -Qm[[1]]];
dTthN = Transpose[Qm].DiagonalMatrix[{0.5, 1., 2.}].Qm;
pars = <|"M" → Mv, "g" → gv, "I" → I0v IdentityMatrix[3],
  "fth" → gam0v IdentityMatrix[3] + epsv dfthN, "fdrag" → xi0v IdentityMatrix[3] +
    epsv dfdrN, "Tth" → epsv dTthN, "Tdrag" → c0v IdentityMatrix[3] + epsv dTdrN|>;
{lam, vec} = Eigensystem[Inverse[pars["Tdrag"]].pars["Tth"]];
e3v = Normalize[vec[[1]]];
lam3 = First@Eigenvalues[dTthN];
Glev = Mv gv (e3v.Inverse[pars["fdrag"]].e3v) /
  (e3v.Inverse[pars["fdrag"]].pars["fth"].e3v);
dAN = (dfthN - gam0v / xi0v dfdrN) / xi0v;
proj[v_, u_] := v - (u.v) u;
Rpdf = c0v Norm[proj[dAN.e3v, e3v]] / Abs[lam3];
Rcorr = c0v Norm[proj[dfthN.e3v, e3v]] / (xi0v Abs[lam3]);
Om3 = Glev lam[[1]];
vbss = Inverse[pars["fdrag"]].(Glev pars["fth"] - Mv gv IdentityMatrix[3]).e3v;
Rexact = Norm[proj[vbss, e3v]] / Abs[Om3] / Sqrt[1 + (Mv Om3 / xi0v) ^ 2];
q0v = Module[{ax = Cross[e3v, Zhat], ang = ArcCos[e3v.Zhat]},
  Join[{Cos[ang / 2]}, Sin[ang / 2] Normalize[ax]]];
pars["gradLnT"] = {0, 0, -Glev};
s0 = Join[{0, 0, 0}, quatR[q0v].vbss, q0v, Om3 e3v];
sol = integrateEOM[pars, s0, 12 Pi / Abs[Om3]];
xs2 = Most@
  Table[sol[t] [[1 ;; 3]], {t, 6 Pi / Abs[Om3], 12 Pi / Abs[Om3], 0.01 (2 Pi / Abs[Om3])}];
ctr = Mean[xs2[[All, 1 ;; 2]]];
rad = Norm[# - ctr] & /@ xs2[[All, 1 ;; 2]]; <|"eps" → epsv, "Omega3" → Om3,
  "R_sim" → Mean[rad], "R_sim spread" → StandardDeviation[rad] / Mean[rad],
  "vZ_sim" → (xs2[[1, 3]] - xs2[[1, 3]]) / (6 Pi / Abs[Om3]), "R_exact" → Rexact,
  "R_corrected (49)" → Rcorr, "R_PDF (49)" → Rpdf, "traj" → xs2|>;
orbitRuns = Table[orbitTest[e, 7], {e, {0.04, 0.02, 0.01}}];

```

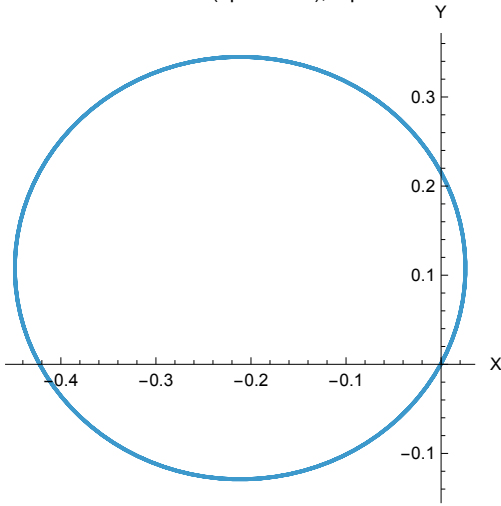
```

Column[
  {Grid[Prepend[Table[{r["eps"], r["R_sim"], r["R_exact"], r["R_corrected (49)"],
    r["R_PDF (49)"], sci[r["R_sim spread"]], sci[r["vZ_sim"]]}], {r, orbitRuns}],
    Style[#, Bold] & /@ {"eps", "R simulated", "R exact quasi-steady",
      "R corrected (49)", "R as printed (49)", "radius spread", "vertical drift"}]],
  Frame → All, Background → {None, {LightGray}}]],
  ListLinePlot[orbitRuns[[2]]["traj"]][[All, 1 ;; 2]], AspectRatio → 1,
  PlotLabel → "simulated orbit (eps = 0.02), top view",
  AxesLabel → {"X", "Y"}, ImageSize → 260], verify["Eqs. (48)-(49)",
  "the steady precession is a horizontal circle with v_Z = 0",
  AllTrue[orbitRuns, #[["R_sim spread"] < 10^-3 && Abs[#[["vZ_sim"]] < 10^-4 &]],
  verify["Eq. (49), corrected", "R_orbit = c0 |Pi_perp df_th
    e3|/(xi0 |lambda3|) converges to the simulation as eps -> 0",
  Module[{errs = Abs[#[["R_corrected (49)"] / #[["R_sim"] - 1] & /@ orbitRuns},
    errs[[3]] < errs[[1]] && errs[[3]] < 0.05],
  "relative errors for eps = 0.04, 0.02, 0.01: " <> StringRiffle[
    sci[Abs[#[["R_corrected (49)"] / #[["R_sim"] - 1] & /@ orbitRuns, ", "]],
  verify["Eq. (49)", "exact quasi-steady radius matches the
    simulation, with the residual shrinking like eps^2",
  Module[{errs = Abs[#[["R_exact"] / #[["R_sim"] - 1] & /@ orbitRuns},
    errs[[3]] < 2*^-4 && errs[[1]] / errs[[3]] > 8],
  "relative errors for eps = 0.04, 0.02, 0.01: " <> StringRiffle[
    sci[Abs[#[["R_exact"] / #[["R_sim"] - 1] & /@ orbitRuns, ", "]], verify["Eq. (49)",
  "R_orbit as printed (with dA containing df_drag) matches the simulated orbit",
  Abs[orbitRuns[[3]]["R_PDF (49)"] / orbitRuns[[3]]["R_sim"] - 1] < 0.05,
  "The printed radius misses by " <> ToString[
    Round[100 Abs[orbitRuns[[3]]["R_PDF (49)"] / orbitRuns[[3]]["R_sim"] - 1]]] <>
  "% at eps = 0.01, and the error does not shrink with
    eps; the corrected radius is within " <>
  num[100 Abs[orbitRuns[[3]]["R_corrected (49)"] / orbitRuns[[3]]["R_sim"] - 1], 2] <>
  "%."], verify["Sec. 2.3, key feature 1",
  "R_orbit is independent of eps (to leading order)",
  Abs[orbitRuns[[1]]["R_sim"] / orbitRuns[[3]]["R_sim"] - 1] < 0.1],
  note["Sec. 2.3, key feature 3", "the hover condition involves only df_th",
  "With the corrected (47), R_orbit = 0 when df_th e3 is
    parallel to e3 (df_th and dT_th share the eigenvector
    e3); the drag tensor plays no role at this order."],
  note["Eq. (42)", "the 'steady' translational balance neglects
    the centripetal acceleration M Omega3 x v",
  "The exact circular solution has |v_perp| reduced by 1/Sqrt[1
    + (M Omega3/xi0)^2]; this is 0(eps^2) here but
    matters for heavy particles or strong asymmetry."]]]

```

| eps  | R simulated | R exact<br>quasi-<br>steady | R corrected<br>(49) | R as<br>printed<br>(49) | radius<br>spread | vertical<br>drift |
|------|-------------|-----------------------------|---------------------|-------------------------|------------------|-------------------|
| 0.04 | 0.22757     | 0.227322                    | 0.224708            | 0.12545                 | 1.94e-9          | 7.73e-6           |
| 0.02 | 0.237027    | 0.236964                    | 0.235223            | 0.126242                | 1.49e-9          | 1.09e-6           |
| 0.01 | 0.241332    | 0.241316                    | 0.240333            | 0.127124                | 1.51e-9          | 1.43e-7           |

simulated orbit (eps = 0.02), top view



✓ **VERIFIED** Eqs. (48)-(49): the steady precession is a horizontal circle with  $v_z = 0$

✓ **VERIFIED** Eq. (49), corrected:  $R_{\text{orbit}} = c_0 |\Pi_{\text{perp}} df_{\text{th}} e_3| / (\xi_0 |\lambda_3|)$  converges to the simulation as  $\varepsilon \rightarrow 0$

relative errors for  $\varepsilon = 0.04, 0.02, 0.01$ : 1.26e-2, 7.61e-3, 4.14e-3

✓ **VERIFIED** Eq. (49): exact quasi-steady radius matches the simulation, with the residual shrinking like  $\varepsilon^2$

relative errors for  $\varepsilon = 0.04, 0.02, 0.01$ : 1.09e-3, 2.64e-4, 6.50e-5

⚠ **DISCREPANCY** Eq. (49):  $R_{\text{orbit}}$  as printed (with  $dA$  containing  $df_{\text{drag}}$ ) matches the simulated orbit

The printed radius misses by 47% at  $\varepsilon = 0.01$ , and the error does not shrink with  $\varepsilon$ ; the corrected radius is within 0.41%.

✓ **VERIFIED** Sec. 2.3, key feature 1:  $R_{\text{orbit}}$  is independent of  $\varepsilon$  (to leading order)

● **NOTE** Sec. 2.3, key feature 3: the hover condition involves only  $df_{\text{th}}$

With the corrected (47),  $R_{\text{orbit}} = 0$  when  $df_{\text{th}} e_3$  is parallel to  $e_3$  ( $df_{\text{th}}$  and  $dT_{\text{th}}$  share the eigenvector  $e_3$ ); the drag tensor plays no role at this order.

● **NOTE Eq. (42):** the 'steady' translational balance neglects the centripetal acceleration  $M \Omega^2 R$

The exact circular solution has  $|V_{\text{perp}}|$  reduced by  $1/\sqrt{1 + (M \Omega^2 R / \xi_0)^2}$ ; this is  $O(\epsilon^2)$  here but matters for heavy particles or strong asymmetry.

### 3. Thermophoretic force on a surface element

We consider a surface element with area  $A$  placed in a thermal field with temperature gradient  $T' > 0$  in the *negative*  $z$ -direction:  $\nabla T = -T' (0, 0, 1) = T' (0, 0, -1)$ , so the gas is hotter at large  $z$  (cold below, hot above) and the thermophoretic force on a sphere is in the  $+z$  direction, opposing gravity. The surface normal in spherical coordinates is  $\hat{n} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$  with tangents  $\hat{t}_1 = (\cos \theta \cos \varphi, \cos \theta \sin \varphi, -\sin \theta)$  and  $\hat{t}_2 = (-\sin \varphi, \cos \varphi, 0)$ .

Following Eqs. (20–21) of Garcia-Ybarra & Rosner (1989), the force on the element is:

$$\frac{1}{A} \begin{pmatrix} F_{t_1} \\ F_{t_2} \\ F_n \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ 0 \\ -\frac{1}{2} \sqrt{\pi/h} \end{pmatrix} - \alpha \begin{pmatrix} -\frac{a}{2} \mathbf{v} \cdot \hat{t}_1 \\ -\frac{a}{2} \mathbf{v} \cdot \hat{t}_2 \\ [2 - a(1 - \pi/4)] \mathbf{v} \cdot \hat{n} \end{pmatrix} + \alpha \frac{3v}{4} \frac{T'}{T} \begin{pmatrix} \frac{a}{2} \sin \theta \\ 0 \\ (2 - a) \cos \theta \end{pmatrix}, \quad (50)$$

where  $\alpha = mN / \sqrt{\pi h}$ . The three terms are hydrostatic pressure, viscous drag, and thermophoretic stress, respectively.

**Independent kinetic-theory derivation of (50).** We derive the free-molecular stress on a flat element from first principles, without using Garcia-Ybarra & Rosner. The gas near the element is described by the Grad (13-moment) distribution carrying a heat flux  $\mathbf{q}$ ,

$$f = N \left( \frac{h}{\pi} \right)^{3/2} e^{-h|\mathbf{c} + \mathbf{v}|^2} \left[ 1 + A(\mathbf{q} \cdot \mathbf{c}) \left( h c^2 - \frac{5}{2} \right) \right], \quad h = \frac{m}{2k_B T},$$

seen from an element moving with velocity  $\mathbf{v}$  relative to the gas, where  $A$  is fixed by requiring the heat-flux moment to equal  $\mathbf{q}$ . Molecules hitting the element (outward normal  $\hat{n}$ ) are re-emitted specularly with probability  $1 - a$ , and diffusely with probability  $a$  from a half-Maxwellian at the gas temperature with the same number flux. The force on the element is the momentum delivered by incoming molecules minus the momentum carried away by outgoing ones. Everything is kept to linear order in  $\mathbf{v}$  and  $\mathbf{q}$ .

```

kinAssume = {hh > 0, nn > 0, mm > 0};
$Assumptions = {hh > 0, nn > 0, mm > 0, kB > 0,
  Tk > 0, kap > 0, nuG > 0, q0 > 0, Rs > 0, rC > 0, LC > 0, 0 ≤ aa ≤ 1};
cvec = {cx, cy, cz}; Vv = {Vx, Vy, Vz}; qv = {qx, qy, qz};
velInt[expr_, czRange_] := Integrate[expr, {cx, -Infinity, Infinity},
  {cy, -Infinity, Infinity}, {cz, Sequence@@ czRange}, Assumptions → kinAssume];
fGrad = nn (hh / Pi) ^ (3 / 2)
  Exp[-hh (cvec + Vv) . (cvec + Vv)] (1 + Ag (qv.cvec) (hh cvec.cvec - 5 / 2));
heatMoment = velInt[(mm / 2) (cvec.cvec) cz
  (fGrad /. {Vx → 0, Vy → 0, Vz → 0, qx → 0, qy → 0}), {-Infinity, Infinity}];
AgSol = First@Solve[heatMoment == qz, Ag];
linearise[expr_] :=
  Normal[Series[expr /. Thread[Join[Vv, qv] → ee Join[Vv, qv]], {ee, 0, 1}]] /. ee → 1;
fLin = linearise[fGrad /. AgSol];
Pin = Table[velInt[mm cvec[[i]] (-cz) fLin, {-Infinity, 0}], {i, 3}];
Gin = velInt[(-cz) fLin, {-Infinity, 0}];
Fspec = {0, 0, 2 Pin[[3]]};
Fdiff = {Pin[[1]], Pin[[2]], Pin[[3]] - Gin mm Sqrt[Pi / (4 hh)]};
Fkin = Simplify[(1 - aa) Fspec + aa Fdiff, kinAssume];
alphaK = mm nn / Sqrt[Pi hh];
Column[{Row[{"Grad coefficient: A = ", TraditionalForm[Ag /. AgSol]}],
  Row[{"stress on the element (tangent x, y; normal z): ", TraditionalForm[
    Collect[Expand[Fkin / alphaK], {Vx, Vy, Vz, qx, qy, qz}, Simplify], " × α"}]}]
Grad coefficient: A =  $\frac{8 \text{ hh}^2}{5 \text{ mm nn}}$ 
stress on the element (tangent x, y; normal z):

$$\left\{ \frac{aa \text{ hh qx}}{5 \text{ mm nn}} - \frac{aa \text{ Vx}}{2}, \frac{aa \text{ hh qy}}{5 \text{ mm nn}} - \frac{aa \text{ Vy}}{2}, -\frac{2 (aa-2) \text{ hh qz}}{5 \text{ mm nn}} + \left(-\frac{\pi aa}{4} + aa - 2\right) \text{ Vz} - \frac{\sqrt{\pi}}{2 \sqrt{\text{hh}}} \right\} \times \alpha$$


```

Now compare term by term with (50), taking  $\mathbf{v}$  in (50) as the element's velocity, which the normal drag term requires. The thermophoretic term of (50) is written for  $\mathbf{q} = \kappa T' \hat{\mathbf{Z}}$ , so that  $\mathbf{q} \cdot \hat{\mathbf{t}}_1 = -q \sin \theta$  and  $\mathbf{q} \cdot \hat{\mathbf{n}} = q \cos \theta$ .

```

coefV[i_, var_] := Simplify[Coefficient[Fkin[[i]], var] / alphaK, kinAssume];
coefQ[i_, var_] := Simplify[Coefficient[Fkin[[i]], var], kinAssume];
pressureKin = Simplify[(Fkin /. Thread[Join[Vv, qv] -> 0]) / alphaK, kinAssume];
thermoTan50 = -alphaK (3 nuSym / 4) (aa / 2) / (kap Tk);
thermoNor50 = alphaK (3 nuSym / 4) (2 - aa) / (kap Tk);
nuSol = First@Solve[thermoNor50 == coefQ[3, qz], nuSym];
Column[{verify["Eq. (50)", "pressure term:  $-\alpha \sqrt{\pi/h}/2 = -N k_B T$ ",
  zeroQ[pressureKin - {0, 0, -Sqrt[Pi / hh] / 2}] &&
  zeroQ[alphaK Sqrt[Pi / hh] / 2 - nn kB Tk /. hh -> mm / (2 kB Tk)]],
verify["Eq. (50)", "normal drag:  $-\alpha [2 - a(1 - \pi/4)] v.n$ ",
  zeroQ[coefV[3, Vz] + (2 - aa (1 - Pi / 4))]], verify["Eq. (50)",
  "tangential drag enters as  $-\alpha(-(a/2) v.t)$ , i.e.  $+(a/2) \alpha v.t$ ",
  zeroQ[coefV[1, Vx] - aa / 2],
  "Kinetic theory gives  $-(a/2) \alpha v.t$ : tangential drag opposes the
    motion, like the normal drag. With the printed sign the tangential
    drag would accelerate the element. The disk and cylinder
    formulas below confirm the kinetic sign."], verify["Eq. (50)",
  "ratio of normal to tangential thermophoretic stress is  $2(2 - a)/a$ ",
  zeroQ[coefQ[3, qz] / coefQ[1, qx] - 2 (2 - aa) / aa]], verify["Eq. (50)",
  "sign of the tangential thermophoretic stress  $+(a/2) \sin \theta$  along  $t_1$ ",
  zeroQ[(thermoTan50 /. nuSol) - coefQ[1, qx]],
  "Kinetic theory: the tangential stress points along the tangential
    heat flux (toward the cold side),  $F_{t1} = -(a/2) \sin \theta \alpha$ 
     $(3 \text{ nu}/4) (T'/T)$ . The printed sign pushes toward the hot side."],
Row[{"identified G-Y coefficient:  $\nu =$  ", TraditionalForm[
  Simplify[nuSym /. nuSol /. hh -> mm / (2 kB Tk), kinAssume]],
  " (with  $\kappa$  the heat conductivity entering  $q = -\kappa \nabla T$ )"}],
note["Eq. (50)", "nu is the kinematic viscosity for a monatomic gas",
  "nu =  $(4/15) \kappa / (N k_B)$ . With the Eucken/Chapman-Enskog value
     $\kappa = (15/4) (k_B/m) \mu$  this is  $\nu = \mu / (m N) = \mu / \rho$ ."],
verify["Sec. 3", "'grad T = -T'(0,0,1), so the gas is hotter at
  large z (cold below, hot above)'", False,
  "T(z) = T0 - T' z decreases upward: the gas is colder at large
    z (hot below, cold above). The stated direction
    of the force (+z, toward cold) is correct."}]

```

✓ **VERIFIED** Eq. (50): pressure term:  $-\alpha \sqrt{\pi/h}/2 = -N k_B T$

✓ **VERIFIED** Eq. (50): normal drag:  $-\alpha [2 - a(1 - \pi/4)] v.n$

⚠ **DISCREPANCY** Eq. (50): tangential drag enters as  $-\alpha(-(a/2) v.t)$ , i.e.  $+(a/2) \alpha v.t$

Kinetic theory gives  $-(a/2) \alpha v.t$ : tangential drag opposes the motion, like the normal drag. With the printed sign the tangential drag would accelerate the element. The disk and cylinder formulas below confirm the kinetic sign.

✓ **VERIFIED** Eq. (50): ratio of normal to tangential thermophoretic stress is  $2(2 - a)/a$

⚠ **DISCREPANCY** Eq. (50): sign of the tangential thermophoretic stress  $(+a/2) \sin \theta$  along  $t_1$

Kinetic theory: the tangential stress points along the tangential heat flux (toward the cold side),  $F_{t1} = -(a/2) \sin \theta \alpha (3 \nu/4) (T'/T)$ . The printed sign pushes toward the hot side.

identified G-Y coefficient:  $\nu = \frac{4 \kappa a p}{15 k_B n n}$

(with  $\kappa$  the heat conductivity entering  $q = -\kappa \nabla T$ )

● **NOTE** Eq. (50):  $\nu$  is the kinematic viscosity for a monatomic gas

$\nu = (4/15) \kappa / (N k_B)$ . With the Eucken/Chapman-Enskog value  $\kappa = (15/4) (k_B/m) \mu$  this is  $\nu = \mu / (m N) = \mu / \rho$ .

⚠ **DISCREPANCY** Sec. 3: ' $\nabla T = -T'(0,0,1)$ , so the gas is hotter at large  $z$  (cold below, hot above)'

$T(z) = T_0 - T' z$  decreases upward: the gas is colder at large  $z$  (hot below, cold above). The stated direction of the force ( $+z$ , toward cold) is correct.

**Integrating over a sphere.** Every surface element of a convex body sees the undisturbed free-molecular gas, so the force on the body is the surface integral of the element stress. With the kinetic stress we recover the two classical free-molecule results for a sphere: Epstein drag with  $\delta = 1 + \pi a/8$ , and Waldmann's thermophoretic force, which is independent of the accommodation coefficient. With the signs printed in (50), the thermophoretic force on a sphere would instead be proportional to  $2 - 2a$  and would vanish for diffuse reflection.

```

nHat[th_, ph_] := {Sin[th] Cos[ph], Sin[th] Sin[ph], Cos[th]};
t1Hat[th_, ph_] := {Cos[th] Cos[ph], Cos[th] Sin[ph], -Sin[th]};
t2Hat[th_, ph_] := {-Sin[ph], Cos[ph], 0};
Ct = coefQ[1, qx]; Cn = coefQ[3, qz];
stressKin[n_, V_, q_] := -alphaK Sqrt[Pi / hh] / 2 n - alphaK
  ((aa / 2) (V - (V.n) n) + (2 - aa (1 - Pi / 4)) (V.n) n) + Ct (q - (q.n) n) + Cn (q.n) n;
stressPDF[n_, t1_, t2_, V_, q_] :=
  Module[{tn = q.n, t1c = q.t1, qq = Sqrt[q.q]}, -alphaK Sqrt[Pi / hh] / 2 n -
    alphaK (- (aa / 2) (V.t1) t1 - (aa / 2) (V.t2) t2 + (2 - aa (1 - Pi / 4)) (V.n) n) + alphaK
      (3 nuSym / 4) (qq / (kap Tk)) ((aa / 2) (-t1c / qq) t1 + (2 - aa) (tn / qq) n) /. nuSol];
sphereForce[stressFn_] := Simplify[Integrate[
  stressFn[th, ph] Rs^2 Sin[th], {th, 0, Pi}, {ph, 0, 2 Pi}], kinAssume && q0 > 0];
Fsph = sphereForce[stressKin[nHat[#1, #2], {0, 0, Vs}, {0, 0, q0}]] &;
FsphPDF = sphereForce[
  stressPDF[nHat[#1, #2], t1Hat[#1, #2], t2Hat[#1, #2], {0, 0, 0}, {0, 0, q0}]] &;
waldmann = 16 Sqrt[Pi] / 15 Rs^2 q0 Sqrt[hh];
epstein = - (4 Pi / 3) Rs^2 (2 alphaK) (1 + Pi aa / 8) Vs;
Column[
  {Row[{"sphere force (kinetic stresses): F_z = ", TraditionalForm[Fsph[[3]]]}],
    verify["Sec. 3 (sphere)", "kinetic stresses give Waldmann's force
      (16 Sqrt[pi]/15) R^2 kappa |grad T| / vbar, independent of a",
      zeroQ[(Fsph[[3]] /. Vs -> 0) - waldmann]], verify["Sec. 3 (sphere)",
      "kinetic stresses give Epstein's drag (4 pi/3) R^2 rho cbar (1 + pi a/8) V",
      zeroQ[(Fsph[[3]] /. q0 -> 0) - epstein]], verify["Eq. (50) (sphere)",
      "the signs printed in (50) reproduce Waldmann's force",
      zeroQ[FsphPDF[[3]] - waldmann], "With the printed tangential sign, F_z = " <>
      ToString[TraditionalForm[Simplify[FsphPDF[[3]] / waldmann]]] <>
      " x Waldmann, which vanishes for full accommodation (a = 1)."]}]
sphere force (kinetic stresses): F_z = - 
$$\frac{\sqrt{\pi} R^2 (5 (\pi aa + 8) \mu \eta Vs - 16 hh q_0)}{15 \sqrt{hh}}$$


```

✓ **VERIFIED Sec. 3 (sphere):** kinetic stresses give Waldmann's force  $(16 \sqrt{\pi/15}) R^2 \kappa |\nabla T| / \bar{v}$ , independent of  $a$

✓ **VERIFIED Sec. 3 (sphere):** kinetic stresses give Epstein's drag  $(4 \pi/3) R^2 \rho \bar{c} (1 + \pi a/8) V$

⚠ **DISCREPANCY Eq. (50) (sphere):** the signs printed in (50) reproduce Waldmann's force

With the printed tangential sign,  $F_z = \backslash!\\(*FormBox[RowBox[{"1", "-", "aa"}], TraditionalForm]) \times$  Waldmann, which vanishes for full accommodation ( $a = 1$ ).

## 4. Thermophoretic force on a disk

A disk of area  $A$  with normal tilted by  $\theta$  from  $\hat{z}$  in the  $x$ - $z$  plane has two surfaces with normals  $\pm \hat{n}$ . Summing both surfaces (pressure cancels, drag and thermophoresis double):

$$\begin{pmatrix} F_{t_1} \\ F_{t_2} \\ F_n \end{pmatrix} = \alpha a \begin{pmatrix} \mathbf{v} \cdot \hat{t}_1 \\ \mathbf{v} \cdot \hat{t}_2 \\ -2[2 - a(1 - \pi/4)] \mathbf{v} \cdot \hat{n} \end{pmatrix} + \alpha \frac{3\nu}{4} \frac{T'}{T} \begin{pmatrix} a \sin \theta \\ 0 \\ 2(2 - a) \cos \theta \end{pmatrix}. \quad (51)$$

```
nD = nHat[thD, 0]; t1D = t1Hat[thD, 0]; t2D = t2Hat[thD, 0];
```

```
Vd = {Vdx, Vdy, Vdz}; qd = {0, 0, q0};
```

```
Fdisk = Simplify[stressKin[nD, Vd, qd] + stressKin[-nD, Vd, qd], kinAssume];
```

```
FdiskLocal = Simplify[{Fdisk.t1D, Fdisk.t2D, Fdisk.nD}];
```

```
F51 =
```

```
{alphaK aa (Vd.t1D), alphaK aa (Vd.t2D), -2 alphaK aa (2 - aa (1 - Pi / 4)) (Vd.nD)} +  
alphaK (3 nuSym / 4) (q0 / (kap Tk)) {aa Sin[thD], 0, 2 (2 - aa) Cos[thD]} /. nuSol;
```

```
Column[{verify["Eq. (51)", "pressure cancels between the two faces",
```

```
zeroQ[Fdisk /. Thread[Join[Vd, {q0}] -> 0]]],
```

```
verify["Eq. (51)", "thermophoretic normal stress doubles to 2(2 - a) cos theta",
```

```
zeroQ[(FdiskLocal[[3]] - F51[[3]]) /. Thread[Vd -> 0]]],
```

```
verify["Eq. (51)", "normal drag is -2 alpha a [2 - a(1 - pi/4)] v.n",
```

```
zeroQ[(FdiskLocal[[3]] - F51[[3]]) /. q0 -> 0],
```

```
"Kinetic result: -2 alpha [2 - a(1 - pi/4)] v.n, the doubled
```

```
normal term of (50). The extra factor a multiplying the
```

```
normal drag in (51) is spurious.", verify["Eq. (51)",
```

```
"tangential terms: +alpha a v.t and +alpha (3 nu/4) (T'/T) a sin theta",
```

```
zeroQ[FdiskLocal[[1]] - F51[[1]]],
```

```
"Kinetic result: -alpha a v.t and -alpha (3 nu/4) (T'/T) a
```

```
sin theta. Both tangential signs inherit the sign slips
```

```
of (50); the magnitudes (doubling) are correct."}]}
```

✓ **VERIFIED** Eq. (51): pressure cancels between the two faces

✓ **VERIFIED** Eq. (51): thermophoretic normal stress doubles to  $2(2 - a) \cos \theta$

⚠ **DISCREPANCY** Eq. (51): normal drag is  $-2 \alpha a [2 - a(1 - \pi/4)] v.n$

Kinetic result:  $-2 \alpha [2 - a(1 - \pi/4)] v.n$ , the doubled normal term of (50). The extra factor  $a$  multiplying the normal drag in (51) is spurious.

⚠ **DISCREPANCY** Eq. (51): tangential terms:  $+\alpha a v.t$  and  $+\alpha (3 \nu/4)(T'/T) a \sin \theta$

Kinetic result:  $-\alpha a v.t$  and  $-\alpha (3 \nu/4)(T'/T) a \sin \theta$ . Both tangential signs inherit the sign slips of (50); the magnitudes (doubling) are correct.

## 5. Thermophoretic force on a cylinder

For a cylinder of radius  $r$  and length  $L$  with its axis tilted at angle  $\theta$  from the  $X$ -axis in the  $X$ - $Z$  plane, and  $\mathbf{v} = (-v, 0, 0)$ , the lab-frame force components are (Garcia-Ybarra & Rosner 1989, Eq. 24):

$$F_X = f \left[ (A \sin^2 \theta + C \cos^2 \theta) v + (B - D) \sin \theta \cos \theta \frac{T'}{T} \right], \quad (52)$$

$$F_Z = f \left[ (A - C) \sin \theta \cos \theta v + (B \cos^2 \theta + D \sin^2 \theta) \frac{T'}{T} \right], \quad (53)$$

where  $f = m N \pi r L / \sqrt{h}$ ,  $A = 2(1 + a(\pi - 2)/8)$ ,  $B = \frac{3v}{2}(1 - a/4)$ ,  $C = a$ ,  $D = 3va/4$ . For  $v = 0$  and  $\theta = 0$  (cylinder axis along  $Z$ ), the levitation force is

$$\vec{F} = \frac{m N \pi r L}{\sqrt{h}} \frac{3v(1 - a/4)}{2} \frac{T'}{T} \hat{Z}, \quad (54)$$

directed upward ( $+\hat{Z}$ ) to oppose gravity. This is proportional to the cylinder volume  $V_c = \pi r^2 L$  in the same way as the sphere result is proportional to  $V_o = \frac{4}{3} \pi r^3$ :

$$\frac{\vec{F}}{V_c} = \left(1 - \frac{a}{4}\right) \frac{\vec{F}_o}{V_o}. \quad (55)$$

The conversion between the Zheng (2002) and Garcia-Ybarra conventions is

$$\frac{2 m N \pi v}{\sqrt{h}} = \frac{\eta \kappa}{\bar{v}}, \quad \bar{v} = \sqrt{2 k_B T / m}, \quad \eta = \frac{16 \sqrt{\pi}}{15} \approx 1.89 \left( \text{free molecular} \right). \quad (56)$$

**Verification of (52)–(56) by integrating the kinetic stress over the lateral surface.** We use an axis  $\hat{a} = (\cos \theta, 0, -\sin \theta)$  (tilted by  $\theta$  below the  $X$ -axis; the opposite tilt sense flips the sign of the two cross terms), cylinder velocity  $(-v, 0, 0)$  and  $\mathbf{q} = \kappa T' \hat{Z}$ . The end caps are neglected ( $L \gg r$ ), as in the notes. The kinetic heat flux is written through  $v$  (from the identification above) and  $T'/T$ . The results fix the prefactor and the angle convention.

```

axC = {Cos[thC], 0, -Sin[thC]};
e1C = {Sin[thC], 0, Cos[thC]};
e2C = {0, 1, 0};
nC[ph_] := Cos[ph] e1C + Sin[ph] e2C;
qOfTau = Simplify[kap Tk tau /. First@Solve[(nuSym /. nuSol) == nuG, kap], kinAssume];
Fcyl = Simplify[Integrate[
    stressKin[nC[ph], {-vC, 0, 0}, {0, 0, qOfTau}] rC LC, {ph, 0, 2 Pi}], kinAssume];
fPDF = mm nn Pi rC LC / Sqrt[hh]; fKin = mm nn Sqrt[Pi] rC LC / Sqrt[hh];
Acoef = 2 (1 + aa (Pi - 2) / 8);
Bcoef = 3 nuG / 2 (1 - aa / 4);
Ccoef = aa;
Dcoef = 3 nuG aa / 4;
F52[f_] :=
    f ((Acoef Sin[thC]^2 + Ccoef Cos[thC]^2) vC + (Bcoef - Dcoef) Sin[thC] Cos[thC] tau);
F53[f_] :=
    f ((Acoef - Ccoef) Sin[thC] Cos[thC] vC + (Bcoef Cos[thC]^2 + Dcoef Sin[thC]^2) tau);
Column[{Row[{"kinetic lateral-surface force: F_X = ", TraditionalForm[
    Collect[Simplify[Fcyl[[1]] / fKin], {vC, tau}, Simplify]], " f, F_Z = ",
    TraditionalForm[Collect[Simplify[Fcyl[[3]] / fKin], {vC, tau}, Simplify]], " f"}],
    verify["Eqs. (52)-(53)", "angular structure and coefficients
    A, B, C, D (with f = mN Sqrt[pi] r L/Sqrt[h])",
    zeroQ[Fcyl[[1]] - F52[fKin]] && zeroQ[Fcyl[[3]] - F53[fKin]]],
    verify["Eqs. (52)-(53)", "prefactor f = m N pi r L / Sqrt[h]",
    zeroQ[Fcyl[[1]] - F52[fPDF]],
    "The kinetic prefactor is f = m N Sqrt[pi] r L/Sqrt[h] =
    alpha pi r L; the printed f is too large by Sqrt[pi]."],
    note["Eqs. (52)-(53)", "sense of the tilt angle",
    "The printed cross terms (A - C) and (B - D) correspond to
    the axis tilted below the X-axis, a = (cos theta, 0, -sin
    theta). For a = (cos theta, 0, +sin theta) both cross terms
    change sign; the notes should state which sense is meant."],
    verify["Eq. (54)", "theta = 0 corresponds to the cylinder axis along Z",
    zeroQ[(axC /. thC -> 0) - {0, 0, 1}],
    "With the angle measured from the X-axis, as stated before (52), theta = 0
    puts the axis along X, perpendicular to the gradient. (54), F = f B
    T'/T, is therefore the force on a horizontal cylinder. A vertical
    cylinder (theta = pi/2) feels F = f D T'/T = f (3 nu a/4) T'/T."], verify[
    "Eq. (54)", "levitation force of a horizontal cylinder is f B T'/T along +Z",
    zeroQ[(Fcyl /. {thC -> 0, vC -> 0}) - {0, 0, fKin Bcoef tau}]]]

```

kinetic lateral-surface force:  $F_X =$

$$vC \left( \frac{1}{4} ((\pi - 6) aa + 8) \sin^2(\text{thC}) + aa \right) - \frac{3}{8} (3 aa - 4) \text{nuG tau} \sin(\text{thC}) \cos(\text{thC})$$

$$f, \quad F_Z = \frac{3}{8} \text{nuG tau} ((4 - 3 aa) \cos^2(\text{thC}) + 2 aa) + \frac{1}{8} ((\pi - 6) aa + 8) vC \sin(2 \text{thC}) \quad f$$

✓ **VERIFIED** Eqs. (52)-(53): angular structure and coefficients A, B, C, D (with  $f = mN \sqrt{\pi r} L/\text{Sqrt}[h]$ )

⚠ **DISCREPANCY** Eqs. (52)-(53): prefactor  $f = m N \pi r L / \text{Sqrt}[h]$

The kinetic prefactor is  $f = m N \sqrt{\pi r} L/\text{Sqrt}[h] = \alpha \pi r L$ ; the printed  $f$  is too large by  $\sqrt{\pi}$ .

● **NOTE** Eqs. (52)-(53): sense of the tilt angle

The printed cross terms (A - C) and (B - D) correspond to the axis tilted below the X-axis,  $a = (\cos \theta, 0, -\sin \theta)$ . For  $a = (\cos \theta, 0, +\sin \theta)$  both cross terms change sign; the notes should state which sense is meant.

⚠ **DISCREPANCY** Eq. (54):  $\theta = 0$  corresponds to the cylinder axis along Z

With the angle measured from the X-axis, as stated before (52),  $\theta = 0$  puts the axis along X, perpendicular to the gradient. (54),  $F = f B T'/T$ , is therefore the force on a horizontal cylinder. A vertical cylinder ( $\theta = \pi/2$ ) feels  $F = f D T'/T = f (3 \vee a/4) T'/T$ .

✓ **VERIFIED** Eq. (54): levitation force of a horizontal cylinder is  $f B T'/T$  along +Z

```

FsphNu = Simplify[waldmann /. q0 -> qOfTau /. {Rs -> rC}, kinAssume];
ratio55 = Simplify[
  ((Fcyl[[3]] /. {thC -> 0, vC -> 0}) / (Pi rC^2 LC)) / (FsphNu / (4 / 3 Pi rC^3)), kinAssume];
eta56 = 16 Sqrt[Pi] / 15;
lhs56 = 2 mm nn Pi nuG / Sqrt[hh]; rhs56 = eta56 kap / Sqrt[2 kB Tk / mm];
kapOfNu = kap /. First@Solve[(nuSym /. nuSol) == nuG, kap];
Column[{Row[{"sphere force in G-Y variables: F_o = ", TraditionalForm[FsphNu]}],
  verify["Eq. (55)", "F/V_c = (1 - a/4) F_o/V_o", zeroQ[ratio55 - (1 - aa / 4)]],
  verify["Eq. (56)",
    "conversion 2 m N pi nu/Sqrt[h] = eta kappa/vbar is dimensionally consistent",
    CompatibleUnitQ[Quantity[1, "Kilograms"] Quantity[1, 1 / "Meters"^3]
      Quantity[1, "Meters"^2 / "Seconds"] Quantity[1, "Meters" / "Seconds"],
      Quantity[1, "Watts" / ("Meters" "Kelvins")] / Quantity[1, "Meters" / "Seconds"]],
    "LHS has units kg/s^2, RHS kg/(s^2 K): a factor T is missing on the right."],
  verify["Eq. (56)",
    "value of the conversion (with kappa from the kinetic identification)",
    zeroQ[Simplify[(lhs56 - rhs56 Tk) /. kap -> kapOfNu /. hh -> mm / (2 kB Tk),
      kinAssume && kB > 0 && Tk > 0]]],
    "Kinetic identification: 2 Sqrt[pi] m N nu/Sqrt[h] = eta kappa T/vbar.
    The printed form has pi instead of Sqrt[pi] and lacks the factor T."],
  verify["Eq. (56), corrected", "2 Sqrt[pi] m N nu/Sqrt[h] = eta
    kappa T/vbar holds exactly", zeroQ[Simplify[
      (2 Sqrt[Pi] mm nn nuG / Sqrt[hh] - rhs56 Tk) /. kap -> kapOfNu /. hh -> mm / (2 kB Tk) ]]],
  verify["Eq. (56)", "free-molecule limit eta = 16 Sqrt[pi]/15 = 1.89 (Waldmann)",
    Abs[N[eta56] - 1.89] < 0.005]]]

```

sphere force in G-Y variables: 
$$F_o = \frac{2 \sqrt{\pi} \text{ mm nn nuG rC}^2 \tau}{\sqrt{hh}}$$

✓ **VERIFIED** Eq. (55):  $F/V_c = (1 - a/4) F_o/V_o$

⚠ **DISCREPANCY** Eq. (56): conversion  $2 \text{ m N } \pi \nu/\text{Sqrt}[h] = \eta \kappa/\text{vbar}$  is dimensionally consistent  
LHS has units  $\text{kg}/\text{S}^2$ , RHS  $\text{kg}/(\text{S}^2 \text{ K})$ : a factor T is missing on the right.

⚠ **DISCREPANCY** Eq. (56): value of the conversion (with  $\kappa$  from the kinetic identification)  
Kinetic identification:  $2 \sqrt{\pi} \text{ m N } \nu/\text{Sqrt}[h] = \eta \kappa T/\text{vbar}$ . The printed form has  $\pi$  instead of  $\sqrt{\pi}$  and lacks the factor T.

✓ **VERIFIED** Eq. (56), corrected:  $2 \sqrt{\pi} \text{ m N } \nu/\text{Sqrt}[h] = \eta \kappa T/\text{vbar}$  holds exactly

✓ **VERIFIED** Eq. (56): free-molecule limit  $\eta = 16 \sqrt{\pi}/15 = 1.89$  (Waldmann)

**Literature cross-check: Garcia-Ybarra & Rosner's sphero-cylinders.** Zheng's review (2002), which the notes cite, summarises Garcia-Ybarra & Rosner (1989) as follows. A long sphero-cylinder aligned

with the gradient drifts 31% faster than a sphere of the same radius; aligned perpendicular it drifts about 6% slower; and at intermediate angles the drift deviates from the gradient by up to 12°. A spherocylinder is a cylinder plus a sphere (its two caps), so the same kinetic stresses give these numbers directly as functions of the aspect ratio  $L/r$ .

```
axialDrag[L_] :=
  Simplify[Integrate[(stressKin[nC[ph] /. thC → Pi / 2, {0, 0, -vC}, {0, 0, 0}][[3]]) rC L,
    {ph, 0, 2 Pi}], kinAssume];
perpDrag[L_] :=
  Simplify[Integrate[(stressKin[nC[ph] /. thC → 0, {0, 0, -vC}, {0, 0, 0}][[3]]) rC L,
    {ph, 0, 2 Pi}], kinAssume];
sphDrag = Simplify[epstein /. {Vs → -vC, Rs → rC}];
FparL = (Fcyl[[3]] /. {thC → Pi / 2, vC → 0}) + FsphNu;
FperpL = (Fcyl[[3]] /. {thC → 0, vC → 0}) + FsphNu;
DparL = axialDrag[LC] + sphDrag; DperpL = perpDrag[LC] + sphDrag;
vRatio[Fx_, Dx_] := Simplify[(Fx / (Dx / vC)) / (FsphNu / (sphDrag / vC)), kinAssume];
parRatio = vRatio[FparL, DparL]; perpRatio = vRatio[FperpL, DperpL];
devAngle[Lr_, av_] :=
  Module[{mu1 = FparL / (DparL / vC), mu2 = FperpL / (DperpL / vC), r12},
    r12 = N[(mu2 / mu1) /. {LC → Lr rC, aa → av} // Simplify];
    NMaxValue[{Abs[ArcTan[r12 Tan[x]] - x], 0 < x < Pi / 2}, x] / Degree];
Column[{Row[{"parallel / sphere velocity ratio: ",
  TraditionalForm[Simplify[parRatio /. LC → Lr rC, kinAssume]]}],
  Grid[Prepend[Table[{Lr, NumberForm[N[parRatio /. {LC → Lr rC, aa → 1}], 3],
    NumberForm[N[perpRatio /. {LC → Lr rC, aa → 1}], 3], NumberForm[
      devAngle[Lr, 1], 3]], {Lr, {5, 10, 15, 20, 50, 1000}}], Style[#, Bold] & /@
    {"L/r", "v_par / v_sphere", "v_perp / v_sphere", "max deviation (deg)"}],
  Frame → All, Background → {None, {LightGray}}, note["Sec. 5 (literature)",
    "the kinetic stresses reproduce the Garcia-Ybarra & Rosner spherocylinder
    trends quoted by Zheng (+31%, about -6%, up to 12 deg)",
    "For full accommodation the parallel speed-up reaches +31% at L/r of about 15
    and tends to pi/8 = +39% for very long rods, the perpendicular slow-down
    is about 7-9%, and the maximum deviation tends to 12 deg. The review
    does not state the aspect ratio or accommodation behind its numbers,
    so this is a consistency check rather than an exact comparison."]]]
parallel / sphere velocity ratio: 
$$\frac{(\pi aa + 8) (3 aa Lr + 8)}{8 (aa (3 Lr + \pi) + 8)}$$

```

| L/r  | v_par / v_sphere | v_perp / v_sphere | max deviation (deg) |
|------|------------------|-------------------|---------------------|
| 5    | 1.23             | 0.935             | 7.72                |
| 10   | 1.29             | 0.926             | 9.37                |
| 15   | 1.31             | 0.922             | 10.1                |
| 20   | 1.33             | 0.921             | 10.5                |
| 50   | 1.37             | 0.917             | 11.3                |
| 1000 | 1.39             | 0.914             | 11.9                |

● **NOTE Sec. 5 (literature):** the kinetic stresses reproduce the Garcia-Ybarra & Rosner spherocylinder trends quoted by Zheng (+31%, about -6%, up to 12 deg)

For full accommodation the parallel speed-up reaches +31% at L/r of about 15 and tends to  $\pi/8 = +39\%$  for very long rods, the perpendicular slow-down is about 7-9%, and the maximum deviation tends to 12 deg. The review does not state the aspect ratio or accommodation behind its numbers, so this is a consistency check rather than an exact comparison.

## 6. Hierarchy of forces and torques for a levitating particle

We now estimate the magnitudes of all four responses — thermophoretic force, thermophoretic torque, gyrophoretic force, and rotational drag torque — for a particle levitated in our thermal cell. The system parameters are: particle radius  $l = 20 \mu\text{m}$ , particle density  $\rho = 1 \text{ g/cm}^3$ , temperature gradient  $|\nabla T| = 200 \text{ K/cm}$ , gas pressure  $P = 5 \text{ Torr}$ , and temperature  $T = 300 \text{ K}$  (rarefied air). The temperature gradient is directed downward ( $\nabla T = -|\nabla T|\hat{z}$ ), so the thermophoretic force on a sphere points upward ( $+\hat{z}$ ) and can balance gravity.

**Gas and particle properties.** The mean thermal speed of air molecules is  $\bar{v} = \sqrt{2k_B T/m} \approx 415 \text{ m/s}$ . At  $P = 5 \text{ Torr}$  the mean free path is  $\lambda \approx 10 \mu\text{m}$ , giving a Knudsen number

$$Kn = \lambda/l \approx 0.5, \quad (57)$$

placing the particle firmly in the transition regime between continuum and free-molecule flow. The particle mass is  $M = \frac{4}{3}\pi l^3 \rho \approx 3.4 \times 10^{-11} \text{ kg}$ .

**Verification of the parameters.** We recompute every number with physical units. The molecular mass of air ( $m = 4.81 \times 10^{-26} \text{ kg}$ , as in the lab's legacy code) and the viscosity of air at 300 K ( $\mu = 1.85 \times 10^{-5} \text{ Pa s}$ ) are the only inputs not stated in the notes. The mean free path uses the standard hard-sphere relation  $\lambda = (\mu/P) \sqrt{\pi k_B T/2m}$ .

```

kBq = Quantity["BoltzmannConstant"];
mAir = Quantity[4.81*^-26, "Kilograms"];
Tq = Quantity[300., "Kelvins"];
Pq = Quantity[5., "Torr"];
ellq = Quantity[20., "Micrometers"];
rhoq = Quantity[1., "Grams" / "Centimeters"^3];
gradTq = Quantity[200., "Kelvins" / "Centimeters"];
kappaq = Quantity[0.026, "Watts" / ("Meters" "Kelvins")];
gq = Quantity[9.81, "Meters" / "Seconds"^2];
muq = Quantity[1.85*^-5, "Pascals" "Seconds"];
eta = 16 Sqrt[Pi] / 15;
vbar = UnitConvert[Sqrt[2 kBq Tq / mAir], "Meters" / "Seconds"];
lam = UnitConvert[muq / Pq Sqrt[Pi kBq Tq / (2 mAir)], "Micrometers"];
Kn = UnitSimplify[lam / ellq];
Mp = UnitConvert[4 / 3 Pi ellq^3 rhoq, "Kilograms"];
close[x_, target_, tol_] := Abs[QuantityMagnitude[x] / target - 1] < tol;
Column[{Grid[{{"vbar", vbar}, {"lambda", lam}, {"Kn", Kn}, {"M", Mp}},
  Frame → All, Alignment → Left],
  verify["Sec. 6", "vbar = Sqrt[2 k_B T/m] = 415 m/s", close[vbar, 415, 0.01]],
  verify["Eq. (57)", "lambda of about 10 um and Kn of about 0.5 at 5 Torr",
    close[lam, 10, 0.1] && Abs[Kn - 0.5] < 0.05],
  verify["Sec. 6", "M = (4/3) pi l^3 rho = 3.4e-11 kg", close[Mp, 3.4*^-11, 0.02]]}]

```

|        |                              |
|--------|------------------------------|
| vbar   | 414.997 m/s                  |
| lambda | 10.2068 μm                   |
| Kn     | 0.510339                     |
| M      | $3.35103 \times 10^{-11}$ kg |

✓ **VERIFIED** Sec. 6:  $v_{\text{bar}} = \sqrt{2 k_B T/m} = 415 \text{ m/s}$

✓ **VERIFIED** Eq. (57):  $\lambda$  of about 10 μm and Kn of about 0.5 at 5 Torr

✓ **VERIFIED** Sec. 6:  $M = (4/3) \pi l^3 \rho = 3.4\text{e-}11 \text{ kg}$

**Dominant response: thermophoretic force.** Using the Waldmann free-molecule result (which is a good approximation even at  $Kn \approx 0.5$ ), the thermophoretic force is

$$F_T = \eta \frac{\kappa}{V} l^2 \left| \nabla T \right| \approx 9.5 \times 10^{-10} \text{ N}, \quad \eta = \frac{16 \sqrt{\pi}}{15} \approx 1.89, \quad (58)$$

where  $\kappa = 0.026 \text{ W/(m K)}$  is the thermal conductivity of air. Gravity gives  $Mg \approx 3.3 \times 10^{-10} \text{ N}$ , so  $F_T/Mg \approx 2.9$ : the thermophoretic force is sufficient to levitate the particle, consistent with the experimental conditions.

**Thermophoretic torque.** The thermophoretic torque arises because different surface elements of a non-spherical body experience different stresses. Its magnitude is set by  $F_T$  acting over a moment arm of order  $l$ :

$$\tau_T \sim \Lambda \left| \nabla \ln T \right| \sim \Gamma l \left| \nabla \ln T \right| = F_T l \approx 1.9 \times 10^{-14} \text{ Nm}, \quad (59)$$

where we used  $\Lambda \sim \Gamma l$  and  $\left| \nabla \ln T \right| = \left| \nabla T \right| / T \approx 66.7 \text{ m}^{-1}$ . The thermophoretic torque is therefore  $l$  times smaller than the thermophoretic force in the sense that  $\tau_T = F_T \cdot l$ , which is the natural geometric relationship between force and torque at the scale of the particle.

```
FT = UnitConvert[eta kappaq / vbar ellq^2 gradTq, "Newtons"];
Mg = UnitConvert[Mp gq, "Newtons"];
gradLnTq = UnitConvert[gradTq / Tq, 1 / "Meters"];
tauT = UnitConvert[FT ellq, "Newtons" "Meters"];
kappaTr = UnitConvert[15 / 4 kBq / mAir muq, "Watts" / ("Meters" "Kelvins")];
Column[{Grid[{{"F_T", FT}, {"M g", Mg}, {"F_T/(M g)", UnitSimplify[FT / Mg]}},
  {"|grad ln T|", gradLnTq}, {"tau_T = F_T l", tauT}],
  Frame -> All, Alignment -> Left], verify["Eq. (58)",
  "F_T = eta (kappa/vbar) l^2 |grad T| = 9.5e-10 N", close[FT, 9.5*^-10, 0.02]],
  verify["Sec. 6", "M g = 3.3e-10 N and F_T/(M g) = 2.9",
  close[Mg, 3.3*^-10, 0.02] && Abs[UnitSimplify[FT / Mg] - 2.9] < 0.05],
  verify["Eq. (59)", "|grad ln T| = 66.7 1/m and tau_T = F_T l = 1.9e-14 N m",
  close[gradLnTq, 66.7, 0.005] && close[tauT, 1.9*^-14, 0.02]],
  note["Eq. (58)", "Waldmann's formula uses the translational conductivity",
  "Waldmann's result is derived for the translational heat
  conductivity. For air, kappa_tr = (15/4) (k_B/m) mu = "<>
  num[QuantityMagnitude[kappaTr], 3] <> " W/(m K), about 25% below the
  total kappa = 0.026 W/(m K), which includes internal molecular degrees
  of freedom. The estimate of F_T should be read as an upper value."]]]
```

|               |                                      |
|---------------|--------------------------------------|
| F_T           | $9.47594 \times 10^{-10} \text{ N}$  |
| M g           | $3.28736 \times 10^{-10} \text{ N}$  |
| F_T/(M g)     | 2.88254                              |
| grad ln T     | 66.6667 per meter                    |
| tau_T = F_T l | $1.89519 \times 10^{-14} \text{ mN}$ |

✓ **VERIFIED Eq. (58):**  $F_T = \eta (\kappa/vbar) l^2 |\nabla T| = 9.5\text{e-}10 \text{ N}$

✓ **VERIFIED Sec. 6:**  $M g = 3.3\text{e-}10 \text{ N}$  and  $F_T/(M g) = 2.9$

✓ **VERIFIED Eq. (59):**  $|\nabla \ln T| = 66.7 \text{ 1/m}$  and  $\tau_T = F_T l = 1.9\text{e-}14 \text{ N m}$

● **NOTE Eq. (58):** Waldmann's formula uses the translational conductivity

Waldmann's result is derived for the translational heat conductivity. For air,  $\kappa_{tr} = (15/4)(k_B/m) \mu = 0.0199 \text{ W/(m K)}$ , about 25% below the total  $\kappa = 0.026 \text{ W/(m K)}$ , which includes internal molecular degrees of freedom. The estimate of  $F_T$  should be read as an upper value.

**Steady-state angular velocity.** A freely suspended non-spherical particle subject to  $\tau_T$  will spin until the rotational drag torque  $\tau_{drag} = C_r \omega$  balances it. Using the free-molecule rotational drag coefficient for a sphere,  $C_r = (8 \pi / 15) (P / \bar{v}) l^4 \approx 4.3 \times 10^{-19} \text{ N m s}$ , the steady-state angular velocity is

$$\omega_{ss} = \frac{\tau_T}{C_r} \sim \frac{F_T l}{C_r} \approx 4.4 \times 10^4 \text{ rad/s} \approx 2 \times 10^{-3} \frac{\bar{v}}{l}. \quad (60)$$

The steady-state spin is much slower than the natural frequency  $\bar{v}/l \approx 2 \times 10^7 \text{ rad/s}$  set by the thermal speed and particle size, confirming the slow-rotation assumption underlying the linear response theory.

**Verification of  $C_r$  from the kinetic stresses of Section 3.** A sphere rotating at  $\omega$  has surface velocity  $\omega \times r$ . The tangential drag stress of the kinetic derivation,  $-(a/2) \alpha V_t$ , then gives a torque  $-C_r \omega$  with  $C_r = \frac{4\pi}{3} a \alpha R^4$ . With  $\alpha = \rho \bar{c}/2$  this is the classical momentum-flux result  $C_r = \frac{2\pi}{3} a \rho \bar{c} R^4$ .

Expressed through  $P$  and  $\bar{v} = \sqrt{2 k_B T / m}$ , it is  $C_r = \frac{8 \sqrt{\pi}}{3} a \frac{P}{\bar{v}} R^4$ . For diffuse reflection ( $a = 1$ ) this is 2.8 times the value used in the notes.

```

torqueSphere =
  Simplify[Integrate[Cross[Rs nHat[th, ph], stressKin[nHat[th, ph], Cross[{0, 0, wz},
    Rs nHat[th, ph]], {0, 0, 0}]] Rs^2 Sin[th], {th, 0, Pi}, {ph, 0, 2 Pi}]];
CrKin = Simplify[-torqueSphere[[3]] / wz];
CrKinPV = Simplify[CrKin /. {nn → Pp / (kB Tk), hh → mm / (2 kB Tk)} /.
  mm → 2 kB Tk / vbarS^2, {Pp > 0, vbarS > 0}];
CrPDF = UnitConvert[8 Pi / 15 Pq / vbar ellq^4, "Newtons" "Meters" "Seconds"];
CrKinQ = UnitConvert[8 Sqrt[Pi] / 3 Pq / vbar ellq^4, "Newtons" "Meters" "Seconds"];
Column[{Row[{"kinetic rotational drag: C_r = ", TraditionalForm[CrKin],
  " = ", TraditionalForm[CrKinPV /. {Pp → "P", vbarS → OverBar["v"]}]}],
  verify["Sec. 6", "C_r = (8 pi/15) (P/vbar) l^4 = 4.3e-19 N m s (arithmetic)",
    close[CrPDF, 4.3*^-19, 0.02]],
  verify["Eq. (60)", "free-molecule rotational drag C_r = (8 pi/15) (P/vbar) l^4",
    zeroQ[(CrKinPV /. aa → 1) - 8 Pi / 15 Pp / vbarS Rs^4],
    "Kinetic theory (diffuse reflection): C_r = (8 Sqrt[pi]/3) (P/vbar) l^4 = "<>
    sci[QuantityMagnitude[CrKinQ]] <> " N m s, a factor "<>
    num[N[(8 Sqrt[Pi] / 3) / (8 Pi / 15)], 3] <> " larger."], verify["Eq. (60)",
    "omega_ss = F_T l / C_r = 4.4e4 rad/s = 2e-3 vbar/l (with the notes' C_r)",
    close[UnitConvert[tauT / CrPDF, 1 / "Seconds"], 4.4*^4, 0.03] &&
    Abs[UnitSimplify[UnitConvert[tauT / CrPDF, 1 / "Seconds"] ellq / vbar] / 2*^-3 - 1] <
    0.1], note["Eq. (60)",
    "with the kinetic C_r the steady spin is 2.8 times slower", "omega_ss = "<>
    num[QuantityMagnitude[UnitConvert[tauT / CrKinQ, 1 / "Seconds"]], 3] <>
    " rad/s. The slow-rotation conclusion is unchanged."]]}
kinetic rotational drag: C_r = 
$$\frac{4 \sqrt{\pi} aa mm nn Rs^4}{3 \sqrt{hh}} = \frac{8 \sqrt{\pi} aa P Rs^4}{3 \bar{v}}$$


```

✓ **VERIFIED** Sec. 6:  $C_r = (8 \pi/15)(P/\bar{v}) l^4 = 4.3e-19 \text{ N m s (arithmetic)}$

⚠ **DISCREPANCY** Eq. (60): free-molecule rotational drag  $C_r = (8 \pi/15)(P/\bar{v}) l^4$   
 Kinetic theory (diffuse reflection):  $C_r = (8 \sqrt{\pi}/3)(P/\bar{v}) l^4 = 1.21e-18 \text{ N m s}$ , a factor 2.82 larger.

✓ **VERIFIED** Eq. (60):  $\omega_{ss} = F_T l / C_r = 4.4e4 \text{ rad/s} = 2e-3 \bar{v}/l$  (with the notes'  $C_r$ )

● **NOTE** Eq. (60): with the kinetic  $C_r$  the steady spin is 2.8 times slower  
 $\omega_{ss} = 15600.0 \text{ rad/s}$ . The slow-rotation conclusion is unchanged.

**Gyrophoretic force at steady-state spin.** The gyrophoretic force (from angular velocity coupling) at steady-state spin  $\omega_{ss}$  is

$$F_\omega \sim \Lambda' \omega_{ss} = \frac{\Lambda}{l \bar{v}} \cdot \frac{F_T l}{C_r} = \frac{F_T l}{\bar{v} C_r} \cdot \frac{F_T}{|\nabla \ln T|} \approx 1.5 \times 10^{-9} \text{ N}. \quad (61)$$

This can be expressed analytically as

$$\frac{F_\omega}{F_T} = \frac{F_T l}{|\nabla \ln T| \bar{V} C_r} = \frac{\eta \kappa T}{\bar{V}} \frac{8\pi}{15 P l} \approx 3 K n \approx 1.6. \quad (62)$$

The gyrophoretic force is therefore comparable to the thermophoretic force for a non-spherical particle spinning at its steady-state rate. This is perhaps surprising: although  $\delta = \omega_{ss} l / \bar{V} \approx 0.002$ , the smallness of  $\delta$  is compensated by the large ratio  $|\mathbf{T}_{th}| / |\mathbf{T}_{drag}| \sim \bar{V} / l^2$ , so that

$F_\omega \sim \delta \cdot (\bar{V} / l) \cdot F_T / |\nabla \ln T| \cdot |\nabla \ln T| \sim F_T$ . The ratio scales as  $Kn$  — it is a rarefaction effect, vanishing in the continuum limit where  $l \gg \lambda$ .

```

Fw = UnitConvert[FT ellq / (vbar CrPDF) FT / gradLnTq, "Newtons"];
ratio62arith = UnitSimplify[FT ellq / (gradLnTq vbar CrPDF)];
ratio62printed = UnitSimplify[eta kappa Tq / vbar (8 Pi) / (15 Pq ellq)];
ratio62fixed = UnitSimplify[eta kappa Tq / vbar 15 / (8 Pi Pq ellq)];
ratioKin = UnitSimplify[FT ellq / (gradLnTq vbar CrKinQ)];
knCoef = Simplify[(eta (15 / 4) (kB / mm) mu Tk / vbarS 15 / (8 Pi Pp ell)) /
  ((mu / Pp Sqrt[Pi kB Tk / (2 mm)]) / ell) /.
  mm -> 2 kB Tk / vbarS^2, {kB > 0, Tk > 0, vbarS > 0}];
Column[{Grid[{"F_omega (61)", Fw},
  {"F_omega/F_T from the first form of (62)", ratio62arith},
  {"(62) as printed: (eta kappa T/vbar) 8 pi/(15 P l)", ratio62printed},
  {"(62) with 15/(8 pi P l)", ratio62fixed}, {"3 Kn", 3 Kn},
  {"F_omega/F_T with the kinetic C_r", ratioKin}], Frame -> All, Alignment -> Left],
verify["Eq. (61)", "F_omega = 1.5e-9 N (with the notes' C_r)",
  close[Fw, 1.5*^-9, 0.05]], verify["Eq. (62)",
  "analytic form (eta kappa T/vbar) 8 pi/(15 P l) equals the first form",
  Abs[ratio62printed / ratio62arith - 1] < 0.01,
  "The printed factor 8 pi/(15 P l) gives "<> num[ratio62printed, 3] <>
  "; substituting C_r = (8 pi/15) (P/vbar) l^4 gives 15/(8
  pi P l) and the stated "<> num[ratio62fixed, 3] <> "."],
verify["Eq. (62)", "F_omega/F_T of about 1.6 and of about 3 Kn",
  Abs[ratio62arith - 1.6] < 0.05 && Abs[ratio62arith / (3 Kn) - 1] < 0.1],
note["Eq. (62)", "the Kn scaling, made explicit",
  "Writing kappa = (15/4) (k_B/m) mu and lambda = (mu/P) Sqrt[pi
  k_B T/2m] turns the corrected (62) into F_omega/F_T = "<>
  ToString[TraditionalForm[knCoef]] <> " Kn = "<> num[N[knCoef], 3] <>
  " Kn. The quoted 3 Kn uses the total conductivity of air."],
note["Eqs. (61)-(62)", "status of the gyrophoretic estimate",
  "Lambda' ~ Lambda/(l vbar) is a dimensional guess, not a derived coupling.
  Within the epsilon-expansion of Section 2, where the notes
  neglect the coupling as O(eps^2), a coupling of order eps times
  a spin of order eps gives F_omega/F_T = O(eps^2). The O(1)
  ratio found here assumes an O(1) asymmetry, eps ~ 1. With the
  kinetic C_r the ratio drops to "<> num[ratioKin, 2] <> "."]]]

```

|                                                   |                                    |
|---------------------------------------------------|------------------------------------|
| F_omega (61)                                      | $1.50738 \times 10^{-9} \text{ N}$ |
| F_omega/F_T from the first form of (62)           | 1.59075                            |
| (62) as printed: (eta kappa T/vbar) 8 pi/(15 P l) | 4.4658                             |
| (62) with 15/(8 pi P l)                           | 1.59075                            |
| 3 Kn                                              | 1.53102                            |
| F_omega/F_T with the kinetic C_r                  | 0.563906                           |

✓ **VERIFIED** Eq. (61):  $F_{\text{omega}} = 1.5\text{e-}9 \text{ N}$  (with the notes'  $C_r$ )

⚠ **DISCREPANCY Eq. (62):** analytic form  $(\eta \kappa T / \bar{v}) 8 \pi / (15 P l)$  equals the first form  
The printed factor  $8 \pi / (15 P l)$  gives 4.47; substituting  $C_r = (8 \pi / 15)(P / \bar{v}) l^4$  gives  $15 / (8 \pi P l)$  and the stated 1.59.

✓ **VERIFIED Eq. (62):**  $F_{\text{omega}}/F_T$  of about 1.6 and of about 3 Kn

● **NOTE Eq. (62):** the Kn scaling, made explicit

Writing  $\kappa = (15/4)(k_B/m) \mu$  and  $\lambda = (\mu/P) \text{Sqrt}[\pi k_B T / 2m]$  turns the corrected (62) into  $F_{\text{omega}}/F_T = \frac{1}{\lambda} \left( \frac{15}{8 \pi P l} \right) \text{Kn} = 2.39 \text{ Kn}$ . The quoted 3 Kn uses the total conductivity of air.

● **NOTE Eqs. (61)-(62):** status of the gyrophoretic estimate

$\Lambda' \sim \Lambda / (l \bar{v})$  is a dimensional guess, not a derived coupling. Within the  $\varepsilon$ -expansion of Section 2, where the notes neglect the coupling as  $O(\varepsilon^2)$ , a coupling of order  $\varepsilon$  times a spin of order  $\varepsilon$  gives  $F_{\text{omega}}/F_T = O(\varepsilon^2)$ . The  $O(1)$  ratio found here assumes an  $O(1)$  asymmetry,  $\varepsilon \sim 1$ . With the kinetic  $C_r$  the ratio drops to 0.56.

**Summary of the hierarchy.** Table 2 summarises all four responses for the present system.

| Response                                                        | Expression                                 | Magnitude                             | Ratio to $F_T$          |
|-----------------------------------------------------------------|--------------------------------------------|---------------------------------------|-------------------------|
| Thermophoretic force $F_T$                                      | $\Gamma  \nabla \ln T $                    | $9.5 \times 10^{-10} \text{ N}$       | 1                       |
| Gravity $Mg$                                                    | $\rho \frac{4}{3} \pi l^3 g$               | $3.3 \times 10^{-10} \text{ N}$       | 0.35                    |
| Thermophoretic torque $\tau_T$                                  | $\Lambda  \nabla \ln T  \sim F_T l$        | $1.9 \times 10^{-14} \text{ N m (†)}$ | 1                       |
| Gyrophoretic force $F_\omega$ (at $\omega_{ss}$ )               | $\Lambda' \omega_{ss} \sim 3 Kn \cdot F_T$ | $1.5 \times 10^{-9} \text{ N}$        | $\sim 3 Kn \approx 1.6$ |
| Rotational drag torque $\tau_{\text{drag}}$ (at $\omega_{ss}$ ) | $C_r \omega_{ss} = \tau_T$                 | $1.9 \times 10^{-14} \text{ N m (†)}$ | 1                       |

Table 2: Estimated magnitudes of the four thermophoretic and drag responses for a levitating particle of generic shape. Parameters:  $l = 20 \mu\text{m}$ ,  $\rho = 1 \text{ g/cm}^3$ ,  $|\nabla T| = 200 \text{ K/cm}$ ,  $P = 5 \text{ Torr}$ ,  $T = 300 \text{ K}$ ,  $Kn \approx 0.5$ . The torque column is normalised by  $l$  for direct comparison with forces. (†) Divided by  $l = 20 \mu\text{m}$  for comparison with forces.\*

Several conclusions follow from this analysis.

- 1. Thermophoretic force and torque are the primary drivers.**  $F_T$  levitates the particle against gravity ( $F_T/Mg \approx 2.9$ ), while  $\tau_T \sim F_T \cdot l$  rotates any non-spherical particle toward a preferred orientation. Both are first-order effects.
- 2. The gyrophoretic force is not negligible.** For a non-spherical particle spinning at steady state, the gyrophoretic force  $F_\omega \sim 3 Kn \cdot F_T$  is of the same order as  $F_T$  in the transition regime ( $Kn \sim 0.5$ ). This coupling between rotation and translation is a rarefaction effect: it vanishes in the continuum limit  $Kn \rightarrow 0$ . For a spherical particle the gyrophoretic coupling vanishes and this term is absent.
- 3. Three small dimensionless parameters govern the dynamics.** The translational drift velocity  $v \sim F_T / \Xi$  satisfies  $v/\bar{v} \approx 4 \times 10^{-4}$ ; the steady spin satisfies  $\omega_{ss}/\bar{v} \approx 2 \times 10^{-3}$ ; and the particle is small compared to the gradient scale,  $l/L_T \approx 1.3 \times 10^{-3}$ . All three confirm the validity of the linear response approximation.

**4. The transition regime requires special treatment.** With  $Kn \approx 0.5$ , neither the continuum ( $Kn \rightarrow 0$ ) nor the free-molecule ( $Kn \rightarrow \infty$ ) limit applies exactly. The full numerical solutions of the Boltzmann equation reviewed in Zheng (2002) are needed for quantitative predictions, particularly for the diagonal tensors  $\mathbf{f}_{\text{th}}$  and  $\mathbf{T}_{\text{drag}}$ . The off-diagonal tensor  $\mathbf{T}_{\text{th}}$  has not been computed for the transition regime.

```
Xi = UnitConvert[4 Pi / 3 ellq ^ 2 (Pq mAir / (kBq Tq)) Sqrt[8 kBq Tq / (Pi mAir)] (1 + Pi / 8),
  "Newtons" "Seconds" / "Meters"];
vDrift = UnitConvert[FT / Xi, "Meters" / "Seconds"];
LT = UnitConvert[Tq / gradTq, "Meters"];
Column[{Grid[{"Epstein drag Xi (diffuse)", Xi},
  {"v = F_T/Xi", vDrift}, {"v/vbar", UnitSimplify[vDrift / vbar]},
  {"omega_ss l/vbar (notes' C_r)", UnitSimplify[tauT / CrPDF ellq / vbar]},
  {"l/L_T", UnitSimplify[ellq / LT]}], Frame -> All, Alignment -> Left],
verify["Table 2", "Mg/F_T = 0.35", Abs[UnitSimplify[Mg / FT] - 0.35] < 0.01],
verify["Sec. 6, conclusion 3",
  "v/vbar of about 4e-4 (order of magnitude, Epstein drag)",
  1*^-4 < UnitSimplify[vDrift / vbar] < 1*^-3,
  "Epstein drag gives v/vbar = "<> sci[UnitSimplify[vDrift / vbar]] <> "."],
verify["Sec. 6, conclusion 3", "omega_ss l/vbar = 2e-3 and l/L_T = 1.3e-3",
  Abs[UnitSimplify[tauT / CrPDF ellq / vbar] / 2*^-3 - 1] < 0.1&&
  Abs[UnitSimplify[ellq / LT] / 1.3*^-3 - 1] < 0.05]]]
```

|                                                    |                                      |
|----------------------------------------------------|--------------------------------------|
| Epstein drag Xi (diffuse)                          | $8.459 \times 10^{-9} \text{ s N/m}$ |
| $v = F_T/X_i$                                      | 0.112022 m/s                         |
| $v/v_{\text{bar}}$                                 | 0.000269934                          |
| $\omega_{\text{ss}} l/v_{\text{bar}}$ (notes' C_r) | 0.002121                             |
| $l/L_T$                                            | 0.00133333                           |

✓ **VERIFIED** Table 2:  $Mg/F_T = 0.35$

✓ **VERIFIED** Sec. 6, conclusion 3:  $v/v_{\text{bar}}$  of about  $4e-4$  (order of magnitude, Epstein drag)  
Epstein drag gives  $v/v_{\text{bar}} = 2.70e-4$ .

✓ **VERIFIED** Sec. 6, conclusion 3:  $\omega_{\text{ss}} l/v_{\text{bar}} = 2e-3$  and  $l/L_T = 1.3e-3$

## 7. Verification summary

Every check made in this notebook, in order. **VERIFIED** statements hold as written. **DISCREPANCY** entries show where the notes need correcting; the corrected form is given in the corresponding verification block. **NOTE** entries record assumptions, ranges of validity and cross-checks.

```
Module[{counts = Counts[$verifications[[All, 3]], rows},
  rows = Table[{Style[v[[1]], 10], Pane[Style[prettyRow[v[[2]]], 10,
    FontFamily → "Source Sans Pro", 330], badge[v[[3]]], {v, $verifications}}];
  Column[{Row[{Style["Total checks: " <> ToString[Length[$verifications]], Bold],
    "      ", badge["VERIFIED"], " ", Lookup[counts, "VERIFIED", 0],
    "      ", badge["DISCREPANCY"], " ", Lookup[counts, "DISCREPANCY", 0],
    "      ", badge["NOTE"], " ", Lookup[counts, "NOTE", 0]}], Grid[
    Prepend[rows, Style[#, Bold] & /@ {"reference", "statement checked", "result"}],
    Frame → All, Alignment → {Left, Center}, Background → {None, {LightGray}},
    ItemSize → {{9, 29, 11}}]], Spacings → 1.5]]
```

**Total checks: 106**      ✓ **VERIFIED** 65      ⚠ **DISCREPANCY** 28      ● **NOTE** 13

| reference        | statement checked                                                                                 | result               |
|------------------|---------------------------------------------------------------------------------------------------|----------------------|
| Eq. (3)          | $R(q)$ is orthogonal with $\det R = 1$ , so $u^b = R^T u$ inverts $u = R u^b$                     | ✓ <b>VERIFIED</b>    |
| Eq. (5)          | antisymmetric part of the drag tensor does no work ( $v.A.v = 0$ )                                | ✓ <b>VERIFIED</b>    |
| Eqs. (4) – (5)   | symmetry of $f_{\text{drag}}$ rests on reciprocity; symmetry of $f_{\text{th}}$ is an assumption  | ● <b>NOTE</b>        |
| Sec. 1.2         | three mutually perpendicular mirror planes force $T_{\text{th}} = 0$                              | ✓ <b>VERIFIED</b>    |
| Eq. (10)         | force rows of (10) reproduce (8)                                                                  | ✓ <b>VERIFIED</b>    |
| Eq. (10)         | torque rows of (10) reproduce (9)                                                                 | ⚠ <b>DISCREPANCY</b> |
| Eq. (11)         | $[f_{\text{th}}] = F L$ : $f_{\text{th}}$ times $\nabla \ln T$ is a force                         | ✓ <b>VERIFIED</b>    |
| Eq. (11)         | $[f_{\text{drag}}] = F T/L$ : $f_{\text{drag}}$ times $v$ is a force                              | ✓ <b>VERIFIED</b>    |
| Eq. (11)         | $[T_{\text{th}}] = F L^2$ : $T_{\text{th}}$ times $\nabla \ln T$ is a torque                      | ✓ <b>VERIFIED</b>    |
| Eq. (11)         | $[T_{\text{drag}}] = F L^2 T/L (= F L T)$ : $T_{\text{drag}}$ times $\omega$ is a torque          | ✓ <b>VERIFIED</b>    |
| Sec. 1.3         | DOF count $6 + 6 + 9 + 6 = 27$                                                                    | ✓ <b>VERIFIED</b>    |
| Eq. (13)         | $M dv/dt = -M g Z - R[f_{\text{th}} R^T \nabla \ln T + f_{\text{drag}} R^T v]$ follows from (8)   | ✓ <b>VERIFIED</b>    |
| Sec. 1.4         | hat map entries are $-\epsilon_{ijk} a_k$                                                         | ✓ <b>VERIFIED</b>    |
| Eq. (14)         | $R \hat{w}^b R^T = \hat{w}^b$ , so $dR/dt = R \hat{w}^b = \hat{w}^b R$                            | ✓ <b>VERIFIED</b>    |
| Sec. 1.4         | ' $\nabla T$ along $-Z$ (hot above cold)'                                                         | ⚠ <b>DISCREPANCY</b> |
| Eq. (16)         | component form of $w \times (I w)$                                                                | ✓ <b>VERIFIED</b>    |
| Eq. (16)         | gyroscopic term vanishes for a sphere and for $w$ along a principal axis                          | ✓ <b>VERIFIED</b>    |
| Eqs. (18) – (19) | quaternion form with $q_0 = \cos(\varphi/2)$ , $q = k \sin(\varphi/2)$ equals the axis-angle form | ✓ <b>VERIFIED</b>    |
| Eq. (20)         | explicit matrix (20) equals (19)                                                                  | ✓ <b>VERIFIED</b>    |
| Eq. (21)         | $dq/dt = Q(q) w^b / 2$ implies $dR/dt = R \hat{w}^b$ , i.e. reproduces (14)                       | ✓ <b>VERIFIED</b>    |
| Eq. (21)         | the quaternion ODE conserves $ q $ exactly ( $q \cdot Q(q) w = 0$ )                               | ✓ <b>VERIFIED</b>    |
| Eqs. (23) – (26) | integration conserves $ q  = 1$ (without explicit renormalisation)                                | ✓ <b>VERIFIED</b>    |
| Eqs. (23) – (26) | $R(q)^T R(q) = 1$ along the solution                                                              | ✓ <b>VERIFIED</b>    |
| Eqs. (23) – (26) | mechanical energy balance with thermophoretic power and dissipation                               | ✓ <b>VERIFIED</b>    |
| Eq. (27)         | generic body: $6 + 6 + 9 + 6 = 27$                                                                | ✓ <b>VERIFIED</b>    |
| Eq. (28)         | three mirror planes: diagonal tensors and $T_{\text{th}} = 0$ , total 9                           | ✓ <b>VERIFIED</b>    |
| Sec. 1.7         | Level 2 heading '(18 DOF)'                                                                        | ⚠ <b>DISCREPANCY</b> |
| Eq. (29)         | uniaxial class ' $C_{\text{inf}v}$ ' has $T_{\text{th}} = 0$ (6 DOF total)                        | ⚠ <b>DISCREPANCY</b> |
| Eq. (29)         | cylinders and spheroids ( $D_{\text{inf}h}$ ): $2 + 2 + 0 + 2 = 6$                                | ✓ <b>VERIFIED</b>    |
| Eq. (30)         | sphere: $1 + 1 + 0 + 1 = 3$                                                                       | ✓ <b>VERIFIED</b>    |
| Sec. 1.7         | chiral bodies allow an antisymmetric part of $f_{\text{th}}$                                      | ● <b>NOTE</b>        |
| Sec. 2 intro     | 'five tensors with up to 24 independent parameters'                                               | ⚠ <b>DISCREPANCY</b> |
| Eq. (31)         | $\gamma_0 = \eta (\kappa/v_{\text{bar}}) l^2$ has the units of $f_{\text{th}}$ (N m)              | ⚠ <b>DISCREPANCY</b> |
| Eq. (31)         | tracelessness of $\delta f_{\text{th}}$ is a definition of                                        | ● <b>NOTE</b>        |

|                         |                                                                                                                          |               |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------|
| Eq. (31)                | $\gamma_0$ (the mean response), not a consequence                                                                        | ● NOTE        |
| Sec. 2.1                | the 'Note on gyrophoretic coupling' sentence is incomplete, and conflicts with Section 6                                 | ● NOTE        |
| Eq. (35)                | $-\epsilon dT_{th}(\nabla \ln T)^b = +G \epsilon dT_{th} R^T Z$ for $\nabla \ln T = -G Z$                                | ✓ VERIFIED    |
| Eq. (37)                | $u = R^T Z$ obeys $du/dt = -w^b \times u$ under (36)                                                                     | ✓ VERIFIED    |
| Eq. (35)                | gyroscopic term is second order for a near-sphere: $w \times (I w)$ with $I = I_0 + \epsilon dI$ and $w = O(\epsilon)$   | ✓ VERIFIED    |
| Eqs. (38) – (39)        | $u = e_k, w^b = (G \epsilon \lambda_k / c_0) e_k$ is a fixed point of (37)                                               | ✓ VERIFIED    |
| Eq. (40)                | lab angular velocity $R w^b_k = (G \epsilon \lambda_k / c_0) Z$ when $R e_k = Z$                                         | ✓ VERIFIED    |
| Eq. (41)                | $\sigma_{kj} = G \epsilon (\lambda_k - \lambda_j) \lambda_j / (I_0 c_0)$ has units of a rate (1/s)                       | ⚠ DISCREPANCY |
| Sec. 2.2                | 'for $\lambda_3 > \lambda_2 > \lambda_1 > 0 \dots$ the stable attractor is then $e_1$ '                                  | ⚠ DISCREPANCY |
| Sec. 2.2 table          | positive spectrum: $e_1$ repeller, $e_2$ saddle, $e_3$ stable attractor                                                  | ✓ VERIFIED    |
| Sec. 2.2                | 'all generic initial conditions converge to eigenmotion 3' (general spectra)                                             | ⚠ DISCREPANCY |
| Sec. 2.2                | positive spectrum: trajectories converge to $e_3$                                                                        | ✓ VERIFIED    |
| Sec. 2.2                | negative spectrum converges to $e_3$ (claim applied beyond the positive case)                                            | ⚠ DISCREPANCY |
| Sec. 2.2                | $V = u^T dT_{th}$ increases monotonically (positive spectrum)                                                            | ✓ VERIFIED    |
| Eq. (42)                | steady state of (13) with $dv/dt = 0$ gives $f_{drag} v^b = G f_{th} u - M g u$                                          | ✓ VERIFIED    |
| Eq. (43)                | first-order expansion of $f_{drag} \wedge \{-1\}(G f_{th} - M g) u$                                                      | ✓ VERIFIED    |
| Eq. (44)                | $v_z = (\gamma_0 G - M g) / \xi_0 + \epsilon G (dA)_{33}$ to first order                                                 | ⚠ DISCREPANCY |
| Eq. (46)                | levitation gradient $G = Mg / (\gamma_0 + \epsilon \xi_0 (dA)_{33})$ to first order                                      | ⚠ DISCREPANCY |
| Eq. (46)                | the two expressions in (46) agree to first order                                                                         | ⚠ DISCREPANCY |
| Eq. (47)                | first-order body-frame velocity at levitation                                                                            | ⚠ DISCREPANCY |
| Eq. (47)                | the two terms of (47) give $u \cdot v^b = 0$                                                                             | ⚠ DISCREPANCY |
| Eq. (47), corrected     | $v^b = (\epsilon G_0 / \xi_0)(df_{th} u - (u \cdot df_{th} u) u)$ is the exact first-order result and is horizontal      | ✓ VERIFIED    |
| Eqs. (48) – (49)        | the steady precession is a horizontal circle with $v_z = 0$                                                              | ✓ VERIFIED    |
| Eq. (49), corrected     | $R_{orbit} = c_0  \Pi_{perp} df_{th} e_3  / (\xi_0  \lambda_3 )$ converges to the simulation as $\epsilon \rightarrow 0$ | ✓ VERIFIED    |
| Eq. (49)                | exact quasi-steady radius matches the simulation, with the residual shrinking like $\epsilon^2$                          | ✓ VERIFIED    |
| Eq. (49)                | $R_{orbit}$ as printed (with $dA$ containing $df_{drag}$ ) matches the simulated orbit                                   | ⚠ DISCREPANCY |
| Sec. 2.3, key feature 1 | $R_{orbit}$ is independent of $\epsilon$ (to leading order)                                                              | ✓ VERIFIED    |
| Sec. 2.3, key feature 3 | the hover condition involves only $df_{th}$                                                                              | ● NOTE        |
| Eq. (42)                | the 'steady' translational balance neglects the centripetal acceleration $M \Omega^2 \times v$                           | ● NOTE        |
| Eq. (50)                | pressure term: $-\alpha \text{Sqrt}[\pi/h]/2 = -N k_B T$                                                                 | ✓ VERIFIED    |
| Eq. (50)                | normal drag: $-\alpha [2 - a(1 - \pi/4)] v \cdot n$                                                                      | ✓ VERIFIED    |
| Eq. (50)                | tangential drag enters as $-\alpha(-(a/2) v \cdot t)$ , i.e. $+(a/2) \alpha v \cdot t$                                   | ⚠ DISCREPANCY |
| Eq. (50)                | ratio of normal to tangential thermophoretic stress is $2(2 - a)/a$                                                      | ✓ VERIFIED    |
| Eq. (50)                | sign of the tangential thermophoretic stress $+(a/2) \sin \theta$ along $t_1$                                            | ⚠ DISCREPANCY |
| Eq. (50)                | $\nu$ is the kinematic viscosity for a monatomic gas                                                                     | ● NOTE        |
| Sec. 3                  | ' $\nabla T = -T'(0,0,1)$ , so the gas is hotter at large $z$ (cold below, hot above)'                                   | ⚠ DISCREPANCY |
| Sec. 3 (sphere)         | kinetic stresses give Waldmann's force $(16 \sqrt{\pi}/15) R^2 \kappa  \nabla T  / \bar{v}$ , independent of $a$         | ✓ VERIFIED    |
| Sec. 3 (sphere)         | kinetic stresses give Epstein's drag $(4 \pi/3) R^2 \rho c_{bar} (1 + \pi a/8) V$                                        | ✓ VERIFIED    |
| Eq. (50) (sphere)       | the signs printed in (50) reproduce Waldmann's force                                                                     | ⚠ DISCREPANCY |
| Eq. (51)                | pressure cancels between the two faces                                                                                   | ✓ VERIFIED    |
| Eq. (51)                | thermophoretic normal stress doubles to $2(2 - a) \cos \theta$                                                           | ✓ VERIFIED    |
| Eq. (51)                | normal drag is $-2 \alpha a [2 - a(1 - \pi/4)] v \cdot n$                                                                | ⚠ DISCREPANCY |

|                      |                                                                                                                                  |               |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------|
| Eq. (51)             | tangential terms: $+\alpha a v \cdot t$ and $+\alpha (3 v/4)(T'/T) a \sin \theta$                                                | ⚠ DISCREPANCY |
| Eqs. (52) – (53)     | angular structure and coefficients A, B, C, D (with $f = m N \sqrt{\pi} r L / \text{Sqrt}[h]$ )                                  | ✓ VERIFIED    |
| Eqs. (52) – (53)     | prefactor $f = m N \pi r L / \text{Sqrt}[h]$                                                                                     | ⚠ DISCREPANCY |
| Eqs. (52) – (53)     | sense of the tilt angle                                                                                                          | ● NOTE        |
| Eq. (54)             | $\theta = 0$ corresponds to the cylinder axis along Z                                                                            | ⚠ DISCREPANCY |
| Eq. (54)             | levitation force of a horizontal cylinder is $f B T'/T$ along +Z                                                                 | ✓ VERIFIED    |
| Eq. (55)             | $F/V_c = (1 - a/4) F_0/V_0$                                                                                                      | ✓ VERIFIED    |
| Eq. (56)             | conversion $2 m N \pi v / \text{Sqrt}[h] = \eta \kappa / v_{\text{bar}}$ is dimensionally consistent                             | ⚠ DISCREPANCY |
| Eq. (56)             | value of the conversion (with $\kappa$ from the kinetic identification)                                                          | ⚠ DISCREPANCY |
| Eq. (56), corrected  | $2 \sqrt{\pi} m N v / \text{Sqrt}[h] = \eta \kappa T / v_{\text{bar}}$ holds exactly                                             | ✓ VERIFIED    |
| Eq. (56)             | free-molecule limit $\eta = 16 \sqrt{\pi}/15 = 1.89$ (Waldmann)                                                                  | ✓ VERIFIED    |
| Sec. 5 (literature)  | the kinetic stresses reproduce the Garcia-Ybarra & Rosner sphero-cylinder trends quoted by Zheng (+31%, about -6%, up to 12 deg) | ● NOTE        |
| Sec. 6               | $v_{\text{bar}} = \text{Sqrt}[2 k_B T/m] = 415 \text{ m/s}$                                                                      | ✓ VERIFIED    |
| Eq. (57)             | $\lambda$ of about 10 $\mu\text{m}$ and Kn of about 0.5 at 5 Torr                                                                | ✓ VERIFIED    |
| Sec. 6               | $M = (4/3) \pi l^3 \rho = 3.4\text{e-}11 \text{ kg}$                                                                             | ✓ VERIFIED    |
| Eq. (58)             | $F_T = \eta (\kappa / v_{\text{bar}}) l^2  \nabla T  = 9.5\text{e-}10 \text{ N}$                                                 | ✓ VERIFIED    |
| Sec. 6               | $M g = 3.3\text{e-}10 \text{ N}$ and $F_T/(M g) = 2.9$                                                                           | ✓ VERIFIED    |
| Eq. (59)             | $ \nabla \ln T  = 66.7 \text{ 1/m}$ and $\tau_T = F_T l = 1.9\text{e-}14 \text{ N m}$                                            | ✓ VERIFIED    |
| Eq. (58)             | Waldmann's formula uses the translational conductivity                                                                           | ● NOTE        |
| Sec. 6               | $C_r = (8 \pi/15)(P/v_{\text{bar}}) l^4 = 4.3\text{e-}19 \text{ N m s}$ (arithmetic)                                             | ✓ VERIFIED    |
| Eq. (60)             | free-molecule rotational drag $C_r = (8 \pi/15)(P/v_{\text{bar}}) l^4$                                                           | ⚠ DISCREPANCY |
| Eq. (60)             | $\omega_{ss} = F_T l / C_r = 4.4\text{e4 rad/s} = 2\text{e-}3 v_{\text{bar}}/l$ (with the notes' $C_r$ )                         | ✓ VERIFIED    |
| Eq. (60)             | with the kinetic $C_r$ the steady spin is 2.8 times slower                                                                       | ● NOTE        |
| Eq. (61)             | $F_{\text{omega}} = 1.5\text{e-}9 \text{ N}$ (with the notes' $C_r$ )                                                            | ✓ VERIFIED    |
| Eq. (62)             | analytic form $(\eta \kappa T / v_{\text{bar}}) 8 \pi / (15 P l)$ equals the first form                                          | ⚠ DISCREPANCY |
| Eq. (62)             | $F_{\text{omega}}/F_T$ of about 1.6 and of about 3 Kn                                                                            | ✓ VERIFIED    |
| Eq. (62)             | the Kn scaling, made explicit                                                                                                    | ● NOTE        |
| Eqs. (61) – (62)     | status of the gyrophoretic estimate                                                                                              | ● NOTE        |
| Table 2              | $Mg/F_T = 0.35$                                                                                                                  | ✓ VERIFIED    |
| Sec. 6, conclusion 3 | $v/v_{\text{bar}}$ of about $4\text{e-}4$ (order of magnitude, Epstein drag)                                                     | ✓ VERIFIED    |
| Sec. 6, conclusion 3 | $\omega_{ss} l / v_{\text{bar}} = 2\text{e-}3$ and $l/L_T = 1.3\text{e-}3$                                                       | ✓ VERIFIED    |

### Principal corrections to the notes.

- Eq. (10).** The block response pairs  $\mathbf{T}_{\text{drag}}$  with  $\mathbf{v}$ . It needs three driving fields,  $((\nabla \ln T)^b, \mathbf{v}^b, \boldsymbol{\omega}^b)$ .
- Directions.**  $\nabla T = - |\nabla T| \hat{\mathbf{z}}$  means hot *below*, cold above (Sections 1.4 and 3).
- Section 1.7.** Three mirror planes give 9 DOF, not 18.  $\mathbf{T}_{\text{th}} = 0$  holds for  $D_{\infty h}$  (cylinders, spheroids), not for  $C_{\infty v}$ , which allows an aligning torque.
- Units of  $\gamma_0$  (Eq. 31) and of the conversion (56).** Both are missing a factor  $T$ :  $\gamma_0 = \eta \kappa l^2 T / \bar{v}$  and  $2 \sqrt{\pi} m N v / \sqrt{h} = \eta \kappa T / \bar{v}$ .
- Stability (Eq. 41).** The rate formula is dimensionally inconsistent. Exact analysis: for a positive-definite  $\delta \mathbf{T}_{\text{th}}$  the largest eigenvector  $\mathbf{e}_3$  is the attractor, as in the table. For a negative spectrum it is  $\mathbf{e}_1$ , and for some mixed spectra no eigenmotion is stable.

- 6. Levitation and orbit (Eqs. 44–49).** The drag anisotropy  $\delta \mathbf{f}_{\text{drag}}$  enters only at  $O(\varepsilon^2)$ . Replace  $\delta \mathbf{A}$  by  $\delta \mathbf{f}_{\text{th}} / \xi_0$ :  $G = Mg / (\gamma_0 + \varepsilon \delta f_{\text{th},33})$ ,  $\mathbf{v}^b = (\varepsilon G_0 / \xi_0) \Pi_{\perp} \delta \mathbf{f}_{\text{th}} \mathbf{u}$ , and  $R_{\text{orbit}} = c_0 | \Pi_{\perp} \delta \mathbf{f}_{\text{th}} \mathbf{e}_3 | / (\xi_0 | \lambda_3 |)$ . The corrected radius is confirmed by direct simulation.
- 7. Surface-element and disk stresses (Eqs. 50–51).** The tangential drag and tangential thermophoretic terms have the wrong sign, and (51) has a spurious factor  $\alpha$  on the normal drag.
- 8. Cylinder (Eqs. 52–54).** The prefactor is  $f = m N \sqrt{\pi} r L / \sqrt{h}$ . The tilt sense should be stated, and  $\theta = 0$  is a horizontal cylinder.
- 9. Rotational drag and gyrophoretic ratio (Eqs. 60, 62).** The free-molecule  $C_r = (8 \sqrt{\pi} / 3) (P / \bar{v}) l^4$  is 2.8 times larger. The analytic form of (62) has  $8 \pi / 15$  inverted.